

THE EPISTEMIC ENGINE

A Deterministic Structural Auditor for Institutional Discourse

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<https://github.com/yacEYX-view/The-Epistemic-Engine>

Abstract

Institutional discourse shapes how knowledge is produced, contested, and legitimized in policy, management, advocacy, and organizational strategy. The structural properties of that discourse, as distinct from its factual claims or moral content, have rarely been subject to systematic, reproducible analysis. The Epistemic Engine addresses this gap through a deterministic, philosophy-grounded auditing system that classifies how institutional text makes its claims rather than what it claims. The system implements a six-layer architecture (structural signal extraction, language-game classification, bias-gate detection, authority-laundering identification, learning-posture derivation, and two-axis rhetorical coordinate placement) as a model-agnostic Python kernel deployed across two AI platforms, OpenAI GPT o3 and Google Gemini.

Validation was conducted against a 69-text corpus comprising 67 documents across six sub-corpora and 2 calibration seeds: 20 SEC regulatory filings serving as baseline, 12 Bureau of Labor Statistics validation documents, 10 energy and oil-market journalism documents, 6 synthetic gate-test cases, 7 externally annotated SemEval hyperpartisan articles, and 12 externally annotated MAFALDA fallacy documents. Scoring used Cohen's Kappa and F1 across two merged dimensions (Classification, Authority Shielding) plus continuous coordinate analysis. Three principal findings emerge. A systematic kernel-violation pattern was documented across three independent corpora: both platforms correctly compute linguistic signal densities at the input layer but do not execute the deterministic co-occurrence logic required to flag authority laundering at Layer 3. A register-mismatch gradient was established empirically: Cohen's κ ranges from 1.000 on synthetic institutional texts to 0.053 on MAFALDA informal argumentation, confirming that institutional structural constructs and general-purpose argumentation fallacies are distinct analytical tasks operating on different lexical mechanisms. A scale contrast with the European Commission Joint Research Centre campaigns demonstrates that the deterministic kernel produces reproducible output on 69 documents with documented failure modes and no inter-annotator calibration requirement, whereas the JRC campaigns required seventy scholars and months of calibration to annotate 2,500 documents with residual expert disagreement. The validation workbook (v2.1, revalidated 2026-03-28) and the kernel-violation registry of 10 confirmed violations are provided as supplementary material.

GitHub repository: <https://github.com/yacEYX-view/The-Epistemic-Engine>

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Declaration

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Abbreviations and Glossary

Key Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
BLS	Bureau of Labor Statistics
CES	Current Employment Statistics (BLS)
CLI	Command-Line Interface
GPT	Generative Pre-trained Transformer (OpenAI model family)
IMA	Inter-Model Agreement
JOLTS	Job Openings and Labor Turnover Survey (BLS)
JRC	Joint Research Centre (European Commission)
KV	Kernel Violation (see Appendix D)
LLM	Large Language Model
MAFALDA	Multi-task Annotated Fallacy Localization and Detection Analysis (NAACL 2024)
NLP	Natural Language Processing
QCEW	Quarterly Census of Employment and Wages (BLS)
RQ	Research Question
SEC	Securities and Exchange Commission
SemEval	Semantic Evaluation (annual NLP competition series)
SPS	School of Professional Studies (NYU)
WBS	Work Breakdown Structure

Glossary of Technical Terms

Term	Definition
Authority laundering	A rhetorical pattern in which authority signals (credentials, institutional names, procedural registers, quantitative framings) are deployed to immunize conclusions from contestation.
Bureaucratic masking	A specific laundering pattern triggered when bureaucratic register cue density and ratio-framing density both exceed the 0.02 co-occurrence threshold.
Coordinate compression	A phenomenon in which one LLM platform produces near-zero values on the engine's coordinate axes while a second platform produces values several orders of magnitude larger, using identical input text and kernel logic.
Defensive routine	Argyris's term for an organizational pattern that protects governing assumptions from examination. In the engine, mapped to the Layer 4 output triggered when any Layer 3 laundering flag fires.
Double-loop learning	Argyris's term for inquiry that revises the governing assumptions themselves rather than adjusting only actions within existing norms. Associated in the engine with open-loop reasoning.
Evasion gate	The first of three Layer 2 bias gates. Fires when evasion_count is at least 2.
Falsifiability	Popper's criterion that genuine empirical claims expose themselves to potential refutation by evidence.
Framing density	The Layer 0 signal measuring the relative frequency of moral or emotional frame-shaping vocabulary per token.
Identity laundering	A laundering pattern triggered when ethos density exceeds 0.03 and either pathos density exceeds 0.02 or evasion_count is at least 2.
Ideological gate	The second Layer 2 bias gate. Fires when normative density exceeds 0.05 or universal density exceeds 0.02, and falsifiability is low.

Term	Definition
Immunizing stratagem	Popper's term for a rhetorical move that insulates a claim from refutation without contributing to its content.
Kernel violation	A deviation from the engine's deterministic specification in which a platform correctly computes signal values at Layer 0 but does not execute the Layer 2 or Layer 3 conditional logic faithfully.
Language game	Wittgenstein's term for a rule-governed communicative practice. In the engine, mapped to the Layer 1 institutional context classifier.
Moral framing gate	The third Layer 2 bias gate. Fires when framing density exceeds 0.02.
Register-mismatch gradient	An empirical finding: Cohen's κ agreement between model implementations tracks the match between input register and the engine's institutional calibration domain, ranging from 1.000 on synthetic institutional text to near zero on informal argumentation.
Scientism	A laundering pattern triggered when ratio density and ethos density both exceed 0.02, representing the simultaneous deployment of quantitative framing and authority cues.

1. Introduction

Large language models and AI-assisted communication tools are now embedded in both the production and consumption of institutional discourse. Organizations use AI to draft, edit, and distribute reports, filings, and press releases; decision-makers use AI to summarize, interpret, and act on those same documents. As the volume of institutional text increases and the cost of producing polished, authoritative-sounding discourse approaches zero, the ability to evaluate that discourse structurally becomes correspondingly more valuable. Decision-makers need to assess not what institutional text claims, but how it makes those claims: whether the claims are open to refutation, whether authority is deployed to foreclose contestation, and whether the communicative posture invites genuine inquiry or protects existing assumptions. This Applied Project documents the design, development, validation, and deployment of the Epistemic Engine, a deterministic philosophy-grounded auditing system that classifies the structural epistemic properties of institutional text. The engine does not fact-check content. It evaluates how institutional text behaves as an epistemic artifact: whether its claims are structured to invite or foreclose refutation, whether it deploys authority to immunize conclusions from contestation, and where its rhetorical posture sits on two axes (an analytic-to-emotive axis and a change-to-preservation axis).

1.1 Sponsor Information and Project Setting

The Applied Project was sponsored by Dr. Andres Fortino, Professor at NYU SPS, and developed at The Digital Forge, an applied research hub that bridges academic research and practical application. The Digital Forge emphasizes intellectual rigor, methodological innovation, and academic accountability through feedback, documented progress, and a focus on publishable results.

The validation set consists entirely of publicly available institutional texts, including Bureau of Labor Statistics releases, White House press statements, SEC filings for Coinbase Global, Super Micro Computer, Tupperware Brands, Kellanova, Tesla, and Microsoft, energy and commodity-market reporting from the *Wall Street Journal* and *Financial Times*, wire syndicate coverage, and externally annotated hyperpartisan articles from SemEval-2023 Task 3 and fallacy documents from the MAFALDA benchmark. No proprietary or restricted materials were used.

The engine's theoretical framework draws on four philosophical traditions. Kierkegaard informs its refusal to claim absolute objectivity, reflected in the preservation of continuous coordinate values and inter-model disagreement rather than binary verdicts. Wittgenstein's language-game theory frames institutional text as rule-governed communication whose meaning depends on context. Popper's falsifiability criterion provides the evaluative standard, while Argyris's organizational learning theory

distinguishes genuine inquiry from defensive routines that shield governing assumptions from scrutiny.

1.2 Overview of the Epistemic Engine

The Epistemic Engine is a six-layer Python pipeline that produces a structured audit report covering five analytical outputs: a bias classification, authority-laundersing flags, a learning-posture label, a testability label, and a two-axis rhetorical coordinate placement. The six layers operate in sequence. Layer 0 converts raw text into 14 signal densities via lexicon-based cue counting normalized per token. Layer 1 classifies the institutional language game being played, drawing on Wittgenstein. Layer 2 applies a strict deterministic gate hierarchy (evasion, ideological, moral framing) implementing Popper's falsifiability criterion. Layer 3 detects three forms of authority laundering (scientism, bureaucratic masking, identity laundering) independently of the bias gate. Layer 4 classifies learning posture as open-loop reasoning or defensive routine, following Argyris. Layer 5 computes a two-axis coordinate on HEAD/HEART and DISRUPTION/STABILITY axes, plotted on an ASCII spectrograph.

All classification logic is deterministic and threshold-based. Given identical signal inputs, the engine always produces identical outputs. This design choice is philosophical as well as operational. A classification system that varies based on model randomness is not an audit; it is an opinion. Popper's demand for reproducible methodology translated

directly into an engineering requirement, and that requirement became the architecture.

The engine's classification kernel, written in Python, is independent of any AI model provider. Current LLM platforms (like OpenAI GPT-3, Google Gemini, and Anthropic Claude) act as interfaces, sending raw text to the deterministic core and formatting the outputs. The LLMs are neither classifiers nor judges. This architectural separation, identified in the validation data and detailed in Section 10 as the kernel-violation pattern, ensures that the deterministic specification is executed accurately when conditional logic is kept separate from the LLM's probabilistic processing.

2. Problem Description and Opportunity

The core problem the Epistemic Engine addresses is the structural opacity of institutional discourse. A systematic gap exists between how institutional text presents itself and how it behaves as an epistemic artifact. Institutional communications routinely present structurally self-protecting claims as though they were evidence-based and open to contestation. The gap widens as AI tools generate and polish an ever-larger share of institutional text. Fluency has improved dramatically; epistemic quality has not.

2.1 The Business Problem in Concrete Terms

The core issue is exemplified by the September 9, 2025 Bureau of Labor Statistics (BLS) Preliminary Benchmark Revision and the White House Press Secretary's statement on

the same data, released the same day. The BLS report, written in a straightforward manner, reveals that the U.S. economy added about 911,000 fewer jobs than initially estimated, describing this as a routine annual revision and emphasizing its preliminary nature. In contrast, the White House statement interprets the same data to support a specific political claim. Although both documents present the same facts, their authoritative tones differ significantly. Structural auditing reveals that the BLS release has no bias, while the White House statement shows evasion and biases indicative of political intent.

2.2 The Scale of the Opportunity

The business problem extends well beyond the BLS paired case. In financial analysis, the gap between SEC Form 10-K and Form 10-Q regulatory disclosures and the discretionary CEO communications that accompany them represents the same structural delta in a high-stakes investment context. The SEC benchmark corpus developed for this project empirically demonstrates the delta across twenty real-world documents covering six corporate events: Coinbase cybersecurity extortion, Super Micro compliance delay, Tupperware solvency crisis, Kellanova merger, Tesla CEO compensation, and Microsoft cyber risk factors. Nine out of ten technical filings cluster in the HEAD+STABILITY quadrant with bureaucratic masking; all ten paired editorial interpretations shift to

HEAD+DISRUPTION without laundering flags, reflecting the market-focused editorial register.

In labor economics reporting, the gap between BLS technical releases and their external interpretations produces the same structural delta in a policy analysis context. The 12-document BLS Validation Corpus assembled for this project includes pairs covering the September 2025 preliminary benchmark revision, the February 2026 final revision, the August 2025 Employment Situation release, the Q4 2025 Employment Cost Index, and the CES Net Birth-Death Model. The paired external interpretations include American Action Forum policy commentary, Econbrowser research discussion, Reuters' market translation of the revision into monthly average job counts, and PIIIE analytical pieces.

In commodity price journalism, the gap between primary WSJ and FT reporting and syndicated wire reconstructions of the same events produces a third domain where the structural delta matters. The 10-document Energy/Oil Market benchmark developed for this project covers five major oil market events: the April 2020 WTI negative-price crash, the 2023 Saudi unilateral production cut, the 2018 Brent crude rally to \$80 per barrel, the 2020 FT negative-price industry analysis, and the November-December 2023 OPEC+ voluntary production cuts. Each primary article is paired with a reconstruction compiled from freely available wire and syndicated sources. The engine detects a structurally distinct laundering pattern in the reconstructions (scientism, a compound of

quantitative framing with institutional authority cues) that is absent from the primary reporting.

The opportunity, therefore, spans three high-stakes institutional discourse domains: regulatory disclosure, labor economics reporting, and commodity price journalism. It also includes the externally annotated hyperpartisan news articles in the SemEval sub-corpus and the general argumentation fallacies in the MAFALDA sub-corpus. Each domain presents its own version of the same structural problem and benefits from the same deterministic auditing capability.

3. Importance of the Project

3.1 Theoretical Significance

The project shows that philosophical frameworks can be effectively turned into computational classification logic while maintaining their conceptual integrity. Popper's falsifiability criterion is transformed into a specification for immunizing strategies, characterized by vague hedging, universal claims immune to counterexamples, and rejection of contrary evidence. This is implemented as a deterministic gate hierarchy. Argyris's concepts of defensive routines and open-loop reasoning are realized as a laundering-flag-based posture classifier, while Wittgenstein's language-game analysis is represented as a signal-combination-based contextual classifier. The direct translation from philosophy to engineering allows for verbatim citations in the engine's

documentation. This work contributes to a key question in computational linguistics about the feasibility of operationalizing structural analytical categories from the humanities, providing affirmative results from a study of 69 documents across six sub-corpora.

3.2 Practical Organizational Value

For decision-makers across sectors, the engine provides a capability that did not previously exist in automated form: the ability to compare documents covering identical factual content along structural epistemic dimensions, producing quantified reproducible delta scores. The engine's use case is not universal text analysis. It is paired-document comparison in institutional contexts where the same underlying facts are reframed for different audiences. The BLS technical release paired with its outside interpretation. The SEC filing paired with its Reuters editorial summary. The WSJ primary report paired with a syndicated wire reconstruction. In each case, the engine provides a quantitative characterization of the structural shift introduced by the second document. For governance and compliance professionals screening institutional communications for disclosure risk, the engine's deterministic architecture is directly aligned with the transparency and interrogability requirements of emerging AI governance frameworks. The European Union Artificial Intelligence Act, adopted in 2024, mandates explainability for AI systems deployed in high-risk contexts. The NIST AI Risk Management Framework

and emerging ISO standards identify epistemic transparency as a core governance requirement. The engine's outputs are fully traceable to specific signal values and specific gate thresholds, which is the design choice most consistent with what these frameworks require of a governance instrument.

3.3 Strategic Positioning

The engine's model-agnostic architecture positions it distinctively in the landscape of AI-assisted text analysis tools. The inter-model agreement data collected in this project provides empirical justification for this design choice: Cohen's κ on the synthetic gate-test corpus is 1.000 (perfect agreement between GPT o3 and Gemini when the kernel can execute), but κ on the Energy validation corpus is 0.302 (fair) and on the BLS corpus is 0.000 (no agreement). The divergence is not random. It reflects each platform's distinct statistical processing patterns at the signal-estimation boundary. Any tool that delegates classification logic to a single LLM will be subject to that platform's biases. The deterministic Python kernel routes around the problem by isolating the classification logic entirely.

4. Alternate Solutions Evaluated

Before settling on the deterministic layered architecture, several alternative approaches were evaluated and rejected. Each is reviewed here, along with the specific reasons it did not meet the evaluation criteria set out in Section 5.

4.1 Automated Fact-Checking Tools

Tools such as ClaimBuster (Hassan et al., 2017) identify specific claims in text and evaluate their correspondence to verified facts. Automated fact-checkers have a clear and legitimate role in journalism and public-information contexts. They do not, however, address the target problem of the Epistemic Engine. The White House statement on the BLS revision contains no factually false claims. Its structural problem is that it presents a predetermined political interpretation as a proven conclusion, an immunizing stratagem that no fact-checker can detect because fact-checking operates at the claim level rather than at the structural level. The engine's target is the compound: a factually accurate wrapper around a self-protecting epistemic posture.

4.2 Sentiment and Tone Analysis Tools

Commercial sentiment analysis tools, including IBM Watson Natural Language Understanding, Google Natural Language AI, Microsoft Azure Text Analytics, and Amazon Comprehend, classify text on positive/negative/neutral axes. Highly evasive institutional language frequently scores as neutral or positive because it uses polished professional vocabulary. Toxicity detectors such as Perspective API and Detoxify operate on a similar principle and face a similar limitation: institutional language designed to obscure accountability is almost never toxic. Readability metrics including Hemingway Editor and Flesch-Kincaid measure surface complexity, which is orthogonal to epistemic

properties. None of these tools captures the compound that the engine targets: high emotional loading combined with authority laundering and immunizing stratagems.

4.3 Machine Learning NLP Classifiers

BERT-based classifiers trained on the MAFALDA and SemEval-2023 Task 3 datasets excel at identifying individual fallacy types but lack the comprehensive structural audit required by governance decision-makers. Three reasons exclude ML classifiers as the primary method. First, their probabilistic outputs cannot be explained, which fails EU AI Act requirements. Second, there is insufficient annotated data for reliable domain-specific ML training, evidenced by MAFALDA's low F1 score of 0.138. Third, ML classifiers create dependencies that this engine aims to avoid. The register-mismatch gradient shows that the engine's analytical categories differ significantly from MAFALDA's labels, indicating that structural auditing and general fallacy detection are distinct tasks. An ML classifier trained on MAFALDA would not work for institutional discourse, and one developed for that purpose would still be limited by MAFALDA's performance ceiling.

4.4 Expanded Human Expert Review

Expert human analysis produces the highest-quality structural classifications. It cannot address the scale problem. The two European Commission Joint Research Centre annotation campaigns provide institutional-scale evidence. The persuasion campaign

required over thirty annotators across six languages with multi-phase quality assurance, weekly calibration meetings, and months of institutional effort (Piskorski et al., 2023). The values campaign required approximately seventy scholars across nine languages (Reitis-Münstermann et al., 2024). Both campaigns produced persistent expert disagreement even under controlled conditions. Both annotated approximately 2,500 documents. Scaling that approach to the volume of institutional text produced daily by corporations, agencies, and policy organizations is operationally infeasible, and the disagreement pattern suggests that scale alone would not resolve the inter-rater variability problem.

4.5 Single-Layer Classifier Designs

A single-layer classifier producing one output (bias classification or laundering flag, but not both) was considered and rejected on analytical grounds. The SEC benchmark corpus and the BLS validation corpus make the case empirically. In the SEC corpus, all twenty documents classify as clean on the bias gate, but the ten technical filings and the ten editorial interpretations separate cleanly on the laundering layer (all ten filings flag bureaucratic masking; all ten editorials do not). A single-layer classifier producing only a bias label would report identical outputs for both groups, missing the entire structural bifurcation. In the synthetic gate-test corpus, document SYN_06 was constructed specifically to test layer independence: it classifies as clean on the bias gate while

triggering all three laundering flags simultaneously. A single-layer design would report clean and miss the triple laundering.

5. Solution Evaluation Criteria

The following criteria were established at the start of the project and used to evaluate candidate architectures.

5.1 Primary Criteria

Determinism and Reproducibility. Given the same raw text and the same lexicon, the engine must always produce the same output, across platforms and across time. An auditing system whose classifications vary based on model randomness is not an audit. This criterion eliminated all ML-based architectures and all architectures that delegated classification logic to LLM prompts.

Transparency and Interrogability. Every output must be traceable to specific signal values, specific gate conditions, and specific layer outputs. A governance decision-maker who sees the engine flag a document must be able to obtain a full deterministic readout explaining which signals crossed which thresholds. This criterion eliminated all black-box classifier designs.

Model-Agnostic Architecture. The tool must not be dependent on any single AI model provider. The performance divergence across platforms documented in Section 8 ($\kappa = 1.000$ between GPT o3 and Gemini on the synthetic corpus, $\kappa = 0.302$ on Energy, $\kappa =$

0.000 on BLS) validates this criterion empirically: no single provider reliably implements the specification across all institutional discourse domains.

Practical Decision-Maker Utility. Outputs must be interpretable and actionable by decision-makers who are not NLP specialists or philosophers. The engine's four-section audit report (snapshot, logic result, spectrograph, ask-for menu) was designed with this criterion in mind.

Validation Against Published Benchmarks. Performance must be evaluable against published peer-reviewed NLP benchmarks. The MAFALDA and SemEval-2023 Task 3 benchmarks were selected as the primary external validation anchors. The project's empirical engagement with 19 externally annotated texts (7 SemEval, 12 MAFALDA) is documented in Section 8.

5.2 Secondary Criteria

Computational efficiency. The engine must produce a full audit in under three seconds on a standard laptop. Achieved: audits complete in under two seconds, with all 14 signal densities computed and all six layers executed.

Multi-platform deployability. The engine must deploy across at least two AI platforms to validate the model-agnostic architecture. Achieved: deployed on OpenAI GPT o3 and Google Gemini formally, and on Anthropic Claude informally for architectural portability confirmation.

Extensibility. The lexicons and the laundering taxonomy must be extensible to support future expansion. Achieved: the lexicon system was expanded from 424 original terms to 1,091 terms across 11 categories during v2.0, and the three-flag laundering taxonomy is designed to accommodate extension toward the 23-technique SemEval-2023 Task 3 set.

6. Selection Rationale

The deterministic layered auditing architecture was selected because it is the only approach that satisfies all five primary evaluation criteria simultaneously.

The determinism criterion eliminates all ML-based architectures. ML classifiers produce probabilistic outputs that are, by design, not reproducible across runs at arbitrary precision. Even with fixed random seeds, the outputs depend on model weights, which are updated as the provider releases new versions. The SEC baseline shift documented in Section 10 (Gemini moving from 20/20 clean on initial validation to 18/20 clean on revalidation 2026-03-28) demonstrates that LLM signal estimation is not temporally stable, confirming that any architecture depending on LLM-based classification inherits that instability.

The transparency criterion eliminates all black-box classifiers. A governance tool whose outputs cannot be decomposed into their inputs cannot meet EU AI Act conformity-assessment requirements or NIST AI RMF measurement functions. The deterministic

layered architecture is the opposite of a black box: every output is traceable to specific cue-count ratios and specific threshold comparisons.

The model-agnostic criterion eliminates architectures whose classification logic is implemented in model-specific prompts or fine-tuned weights. The Python kernel implementation allows any LLM to serve as an interface wrapper while keeping the classification logic identical across platforms. The kernel-violation finding documented in Section 10 is the direct empirical validation of this choice: both GPT o3 and Gemini exhibit systematic violations of the deterministic specification at Layer 3, and the only reliable way to ensure correct co-occurrence detection is to execute the conditional logic in the Python kernel, independent of the LLM.

The decision-maker utility criterion eliminates architectures whose outputs require specialist interpretation. The engine's ASCII spectrograph and its plain-language audit snapshot are readable by decision-makers without NLP training or philosophical background. The ask-for menu exposes the deterministic readout to users who want to audit the audit itself.

The benchmark validation criterion anchors the engine's analytical categories to published NLP research traditions. The MAFALDA taxonomy mapping, SemEval Task 3 genre classification mapping, and JRC persuasion-framework mapping are documented in Section 7 and revisited in Section 8 with empirical measurements.

No single alternative architecture considered during project planning satisfied more than three of the five criteria. The deterministic layered design satisfies all five and, as the validation results show, does so with documented failure modes and a clear path to improvement.

7. Approach and Methodology

7.1 Research Questions

The project was organized around two research questions:

Research Question 1. Can a deterministic, philosophy-grounded auditing architecture reliably classify the structural epistemic properties of institutional discourse across multiple domains?

Research Question 2. Do the engine's outputs provide practically useful information to decision-makers evaluating institutional communications, including the ability to detect structurally distinct laundering patterns across document types?

7.2 System Architecture

The engine implements a six-layer deterministic pipeline. Each layer addresses a distinct analytical question. All classification logic is threshold-based with no stochastic components. Given identical signal inputs, the engine always produces identical outputs. An important architectural distinction applies throughout: the classification logic at Layers 1 through 5 is fully deterministic, but the signal-estimation step at Layer 0

depends on how the underlying AI platform tokenizes and processes text, producing model-dependent variation at the input boundary. This is where the kernel violations documented in Section 10 originate. The platforms compute the signal numbers correctly, then do not apply the conditional logic faithfully.

Table 7-2 summarizes the six layers, the analytical question each addresses, and the key output produced.

Table 7-2. System Architecture Layers

Layer	Name	Question Answered	Key Output
0	Structural Extraction	What signals does this text emit?	14 signal densities; falsifiability flag
1	Wittgenstein Language Game	What institutional game is being played?	Game type: policy defense, PR statement, scientific justification, polemic, corporate risk disclosure, editorial advocacy, legislative persuasion
2	Popper Bias Gate	Are claims open to refutation or self-protecting?	Classification: clean / evasion / ideological / moral_framing; Popper marker: falsifiable_claim_structure / immunizing_stratagem
3	Authority Laundering	Is authority being laundered through form or identity?	Flags: scientism / bureaucratic_masking / identity_laundering
4	Argyris Learning Posture	Does the discourse invite learning or protect assumptions?	Learning posture: open_loop_reasoning / defensive_routine
5	Coordinate Spectrograph	Where does this discourse sit in affect-cognition and change-stability space?	X, Y coordinates; intensity 0 to 10; quadrant Q1 to Q4 or CLEAN; ASCII spectrograph

7.2.1 Layer 0: Structural Extraction

Layer 0 converts raw text into 14 signal densities via lexicon-based cue counting normalized per token count. The signal lexicons were expanded from 424 original terms to 1,091 terms across 11 categories during v2.0 using a function-first, domain-agnostic methodology. A word belongs in a lexicon if and only if it performs that rhetorical function in institutional discourse, regardless of domain. Signals are organized into four functional groups: five gate inputs, four coordinate inputs, four laundering inputs, and one reasoning input. The gate inputs are `evasion_count`, `falsifiability_low`, `normative_density`, `universal_density`, and `framing_density`. The coordinate inputs are `head_density`, `heart_density`, `disruption_density`, and `stability_density`. The laundering inputs are `ratio_density`, `ethos_density`, `lex_density`, and `pathos_density`. The reasoning input is `logos_density`. The `falsifiability_low` flag is set when any of four conditions hold: two or more vagueness cues, one or more unfalsifiable cues, one or more shifting-criteria cues, or two or more totalizing group-attribution patterns.

7.2.2 Layer 1: Wittgenstein Language Game

Layer 1 classifies the text's contextual role from dominant signal combinations. Seven language-game categories are defined: policy defense, PR statement, scientific justification, polemic, corporate risk disclosure, editorial advocacy, and legislative persuasion. This classification matters because the same vocabulary performs different

functions in different language games. The word 'compliance' in an SEC filing performs bureaucratic register; the same word in an editorial performs authority invocation.

7.2.3 Layer 2: Popper Bias Gate Hierarchy

The bias gate implements Popper's falsifiability criterion as a fixed-order hierarchy.

Gates are evaluated in strict sequence and the first gate that fires determines classification. The evasion gate fires when `evasion_count` is at least 2; the threshold is set at 2 rather than 1 because a single evasion cue can reflect legitimate emphasis. The ideological gate fires when `normative_density` exceeds 0.05 or `universal_density` exceeds 0.02, and `falsifiability_low` is true. The moral framing gate fires when `framing_density` exceeds 0.02. If no gate fires, the classification is clean. The derived Popper marker is `immunizing_stratagem` if any gate fires, and `falsifiable_claim_structure` otherwise.

7.2.4 Layer 3: Authority Laundering

Three laundering flags are evaluated independently of the bias gate. Scientism fires when `ratio_density` and `ethos_density` both exceed 0.02, representing the compound of quantitative framing with authority cues. Bureaucratic masking fires when `lex_density` and `ratio_density` both exceed 0.02, representing the compound of procedural register with quantitative framing. Identity laundering fires when `ethos_density` exceeds 0.03 and either `pathos_density` exceeds 0.02 or `evasion_count` is at least 2.

Layer 3 operates independently of Layer 2. A document can classify as clean at the bias gate while simultaneously triggering one or more laundering flags. The synthetic gate-test case SYN_06 validates this independence: it produces a clean bias classification while triggering all three laundering flags simultaneously (scientism, bureaucratic masking, and identity laundering).

Layer 3 is also where the kernel violation pattern documented in this project concentrates. Both LLM platforms produce signal densities that satisfy the co-occurrence conditions (for example, `lex_density` of 0.12 and `ratio_density` of 0.05, both exceeding 0.02) but report `bureaucratic_masking` as false. The specification is unambiguous: if both densities exceed threshold, the flag fires. The platforms do not execute this conditional. This disconnect is not a specification error or an edge case. It is a systematic execution gap at the Layer 0 to Layer 3 boundary, documented across synthetic, SEC, and MAFALDA corpora in the kernel violation registry.

7.2.5 Layer 4: Argyris Learning Posture

Layer 4 derives directly from Layer 3. If any laundering flag is present, the text is classified as `defensive_routine`, compatible with Argyris's single-loop learning (the text protects its conclusions from evidential revision). If no laundering flag is present, the text is classified as `open_loop_reasoning`, compatible with double-loop learning (the text's claim structures are open to revision).

7.2.6 Layer 5: Coordinate Spectrograph

Layer 5 computes two continuous axes. The Y axis is (head_density minus heart_density) times 10, where positive values are analytic and negative values are emotive. The X axis is (disruption_density minus stability_density) times 10, where positive values are change-oriented and negative values are preservation-oriented. Both axes are clamped to the range [-10, +10]. Intensity is computed as the Euclidean magnitude divided by $\sqrt{200}$, times 10, yielding a value normalized to [0, 10]. Quadrant assignment follows sign combinations: Q1 is HEAD+STABILITY, Q2 is HEAD+DISRUPTION, Q3 is HEART+STABILITY, Q4 is HEART+DISRUPTION, and CLEAN is returned when both axes are below 3 in absolute value (insufficient signal magnitude for reliable placement). The spectrograph is rendered as a five-by-five ASCII grid for inclusion in audit reports.

7.3 Validation Framework

The validation framework, designated v2.1 and revalidated on 2026-03-28, consolidates 69 texts comprising 67 documents across six sub-corpora and 2 calibration seeds. The analytical structure uses two merged dimensions (Classification, combining gate output and testability; Authority Shielding, combining laundering flags and learning posture) plus continuous coordinate analysis. Cohen's κ is computed on the two dichotomous merged dimensions; coordinate analysis is reported as mean absolute difference rather than κ because the values are continuous. The SEC benchmark serves as the baseline corpus; all other corpora serve validation roles. Table 7-1 presents the corpus structure.

Table 7-1. Validation Corpus Structure (69 documents)

Corpus	Role	n	Function and Revalidation Status
SEC Benchmark	BASELINE	20	Ground-truth anchor: 10 technical filings plus 10 editorial interpretations covering Coinbase, Super Micro (three filings), Tupperware, Kellanova/Mars (two), Tesla (two), and Microsoft. Revalidation 2026-03-28 showed Gemini moving from 20/20 clean to 18/20 clean.
BLS Validation	VALIDATION	12	Labor-economics register generalization. Six document pairs covering the September 2025 CES preliminary benchmark, February 2026 final revision, August 2025 Employment Situation, Q4 2025 Employment Cost Index, CES Net Birth-Death Model, and BLS reporting framework pieces. Outside interpretations from American Action Forum, Econbrowser, PIIIE, and Reuters.
Energy/Oil Market	VALIDATION	10	Commodity-price journalism. Five pairs of WSJ and FT primary reporting with syndicated wire reconstructions covering the April 2020 WTI negative-price crash, the 2023 Saudi unilateral cut, the 2018 Brent rally to 80 dollars, the 2020 FT negative-price industry analysis, and the November-December 2023 OPEC+ voluntary cuts.
Synthetic Gate Tests	VALIDATION	6	One document per classification pathway. SYN_01 (evasion), SYN_02 (ideological), SYN_03 (moral_framing), SYN_04 (clean, no laundering), SYN_05 (clean plus editorial register boundary), SYN_06 (clean plus all three laundering flags). Both models achieved 6/6 on gate classification.
SemEval Hyperpartisan	VALIDATION	7	Externally annotated articles from the SemEval-2023 Task 3 hyperpartisan corpus. Independent ground truth produced without engine involvement.

Corpus	Role	n	Function and Revalidation Status
MAFALDA External	VALIDATION	12	Blind-scored fallacy corpus from Helwe et al. (2024). Independent ground truth. Establishes the register-mismatch gradient documented in Section 8.
Calibration Seeds	CALIBRATION	2	Original synthetic anchor cases (EE_0001, EE_0002). Excluded from the six-corpus count cited in the abstract; these documents serve as calibration anchors, not validation units.

Two sub-corpora provide ground truth produced entirely without engine involvement.

The 7-article SemEval hyperpartisan set was annotated by the SemEval-2023 Task 3 organizers. The 12-text MAFALDA sample was annotated by expert human raters per Helwe et al. (2024). Together, these 19 documents constitute the only portion of the validation framework where the reference classifications are fully independent of the engine's own outputs. This matters for interpreting the inter-model agreement statistics: on the 19 externally annotated documents, κ measures agreement with independent ground truth; on the other corpora, κ measures agreement with reference classifications that were adjudicated using the engine's deterministic specification.

7.3.1 Platform Selection

GPT o3 and Gemini were selected as the two formal validation platforms to enable inter-model agreement measurement via Cohen's κ against documents with socially designated classification status (for example, SEC corporate filings as institutionally neutral baseline documents). Informal testing on a third platform, Anthropic Claude,

confirmed architectural portability. The two-platform design supports a controlled A/B comparison of each model's original annotation tendencies against the deterministic kernel specification.

7.3.2 Experimental Conditions

Testing was conducted under three conditions. First, the primary BLS paired test comparing the BLS technical release with the White House statement covering identical data. Second, the broader 22-document BLS Validation Corpus scored independently by both platforms against the adjudicated reference classifications. Third, the six synthetic gate-test cases constructed to instantiate every classification pathway, scored for unit-test correctness of the kernel across every combination of bias gate and laundering flag outcomes.

7.3.3 Benchmark Engagement

The validation strategy was extended beyond qualitative taxonomy mapping to empirical measurement against published NLP benchmarks. Twelve MAFALDA texts were selected to cover the taxonomy's three top-level categories (Pathos, Ethos, Logos) and processed blind by both engine implementations. The MAFALDA gold-standard labels were assigned by trained human annotators without any engine involvement. The taxonomy mapping was locked before the blind run with no post-hoc adjustments. MAFALDA's ad hominem maps to the evasion gate. False dilemma maps to the ideological gate. Appeal to worse problems and appeal to ridicule map to the moral

framing gate. Logos structural defects (false causality, circular reasoning, hasty generalization) map to clean. Seven SemEval-2023 Task 3 hyperpartisan articles were processed similarly against the task organizers' annotations.

8. Project Objectives and Metrics

8.1 Project Goal

The project goal was to build a working proof of concept demonstrating that the structural epistemic properties of institutional discourse can be measured, classified, and reported automatically, reproducibly, and without dependence on any single AI model provider.

8.2 Project Objectives and Results

Seven objectives were specified at project outset and approved by the sponsor. Table 8-1 presents the objective list along with the measured result against each objective.

Table 8-1. Results Against Project Objectives

#	Objective	Result
1	Implement a six-layer deterministic architecture in Python with a complete executable CLI	Met. All six layers execute without error; outputs produced in under two seconds. Full CLI deployed as <code>epistemic_engine.py</code> v2.0.0.
2	Achieve at least 70 percent primary classification match rate against expert-coded reference classifications	Met. Overall match rate of approximately 75 percent. Match rate exceeds 80 percent on documents with strong structural signals and approximately 70 percent on near-threshold cases.

#	Objective	Result
3	Validate 100 percent of synthetic reference gate seed cases	Met. All six synthetic gate-test cases (SYN_01 through SYN_06) produced correct bias-gate classification on both GPT o3 and Gemini implementations. Layer 3 laundering execution revealed kernel violations (see Objective 6 and Section 10).
4	Deploy classification logic across at least two AI platforms	Met and exceeded. Three platforms deployed: GPT o3 and Gemini formally, Claude informally. Model-agnostic Python core operates identically across all three.
5	Construct a reference validation workbook with MAFALDA-mapped annotations for at least seven test cases	Met and exceeded. Workbook v2.1 (revalidated 2026-03-28) contains 69 texts (67 documents across six sub-corpora plus 2 calibration seeds) with 53 MAFALDA-aligned annotations, Cohen's κ scoring across two merged dimensions, and a kernel violation registry documenting 10 confirmed violations.
6	Build an automated test suite with at least 50 assertions covering all six layers	Met and exceeded. 89 pytest assertions, all passing in v2.0.0. The test suite executes the Python kernel directly and does not depend on LLM signal estimation, which is why it passes despite the kernel violations observed on LLM platforms.
7	Produce a complete structured audit report for every test document	Met. All 69 documents produced complete four-section audit outputs (audit snapshot, logic result, spectrograph, ask-for menu).

8.3 Inter-Model Agreement Across All Corpora

The Inter-Model Agreement sheet of the v2.1 workbook presents Cohen's κ and F1 for each validation corpus across the two merged dimensions. Table 8-2 reproduces the classification-dimension results. Authority Shielding results are reported in the narrative that follows.

Table 8-2. Inter-Model Agreement, Classification Dimension

Corpus	n	Obs Agr	κ	Strength	F1 GPT	F1 Gem	Key Finding
SEC Baseline	20	0.800	0.375	Fair	0.850	0.900	Baseline anchor; Gemini 18/20 post-revalidation
Synthetic	6	1.000	1.000	Perfect	1.000	1.000	Gate 6/6 both models
SemEval	7	0.857	0.462	Moderate	0.857	1.000	Article 1 gate divergence only
Energy	10	0.700	0.302	Fair	0.700	1.000	Scientism pattern confirmed
BLS	12	0.833	0.000	No agr.	0.303	1.000	GPT all-clean majority bias
MAFALDA	12	0.750	0.053	Slight	0.000	0.053	Register mismatch

The κ gradient across corpora is itself informative. On synthetic institutional text with strong signals, both models achieve perfect agreement. On externally annotated news articles (SemEval), agreement is moderate. On institutional validation corpora (Energy, BLS), it degrades to fair or zero. On informal argumentation (MAFALDA), it is near zero. This gradient tracks the degree to which the input register matches the engine's institutional calibration domain and the degree to which each LLM's statistical processing can reliably estimate the 14 signal densities that feed the deterministic

kernel. The gradient is the second principal finding of this project, discussed further in Section 13.

On the Authority Shielding dimension, two patterns emerge. On the Energy corpus, κ equals 0.400 (Fair) with GPT F1 equal to 0.545 and Gemini F1 equal to 1.000, driven by Gemini's detection of scientism on WSJ primary reporting that GPT does not flag. On the BLS corpus, κ equals 0.000 (no agreement) with GPT F1 of 0.000, driven by GPT's conservative laundering detection: GPT reports no laundering on any BLS technical release, while Gemini detects bureaucratic masking on BLS_B01 and BLS_B02 and scientism on BLS_B02 and BLS_B04. The Synthetic corpus produces κ equal to 0.455 (Moderate) for Authority Shielding, driven by the Layer 3 kernel violations documented in Section 10.

8.4 Kernel Violation Registry

The kernel violation registry documents deviations from the deterministic specification in which an LLM platform correctly computes signal values at Layer 0 but does not execute the Layer 2 or Layer 3 conditional logic faithfully. Ten confirmed violations were documented in the v2.1 revalidation across three independent corpora. Table 8-3 summarizes the violations by corpus and layer.

The full registry is provided in Appendix D.

Table 8-3. Kernel Violation Registry Summary

Corpus	Documents	Violation Count	Description
Synthetic	SYN_01, SYN_02, SYN_05, SYN_06	5 violations (KV-01 through KV-04, KV-10)	GPT under-fires bureaucratic_masking on SYN_01, SYN_02, SYN_06 despite lex and ratio densities both exceeding 0.02. Gemini under-fires both scientism and bureaucratic_masking on SYN_05 despite ratio, ethos, and lex densities all exceeding threshold.
SEC	Doc 3.5, Doc 4.5, Doc 8.5	3 violations (KV-05, KV-06, KV-07)	GPT over-fires identity_laundering on docs 3.5 and 4.5 with ethos densities below the 0.03 threshold. GPT over-fires ideological on doc 8.5 with normative and universal densities both below their respective gate thresholds.
MAFALDA	MAF_03, MAF_11	2 violations (KV-08, KV-09)	GPT under-fires moral_framing gate despite framing_density exceeding 0.02 on both documents.

Layer 2 violations (gate classification departing from signal data) appear only on GPT and only on editorial-register text. Layer 3 violations (laundering co-occurrence conditions met but flag not fired) appear on both platforms and across three corpora. The Layer 3 pattern is the more consequential finding because it is universal: both platforms, across three independent corpora, compute the signal densities correctly and then do not evaluate the conditional logic. The violation pattern is structured rather than random. This is the first principal finding of the project.

9. Risk Analysis

A risk register was established at project outset to identify technical, validation, platform, and organizational risks. Each identified risk was assigned a mitigation strategy, and the register was reviewed at each major phase transition. Table 9-1 presents the risk register along with the outcome for each risk at project close.

Table 9-1. Risk Register and Mitigations

Risk	Mitigation Strategy	Outcome
Technical: Lexicon-based signal computation may be insufficient to classify structural properties reliably	Validate lexicon approach against published benchmarks before committing to it; iterate lexicons based on empirical results	Partially materialized. The register-mismatch finding on MAFALDA shows that lexicon density alone is insufficient for informal argumentation. Mitigation worked for institutional discourse; the limitation is documented and bounded.
Technical: Threshold sensitivity near classification boundaries may produce classification changes that do not reflect genuine structural differences	Document near-threshold cases qualitatively and flag them as requiring human review	Did materialize. The SEC baseline shift from 20/20 clean to 18/20 clean on revalidation (Gemini) is a direct instance. Flagged for v2.1 threshold recalibration.
Validation: Quality and coverage of the validation workbook may be insufficient for inter-rater reliability claims	Expand corpus from 7 seed texts to a multi-corpus set with formal κ and F1 scoring	Fully mitigated. Corpus expanded to 69 texts across six sub-corpora plus two calibration seeds with complete IMA scoring.

Risk	Mitigation Strategy	Outcome
Validation: Gold-standard circularity: reference classifications adjudicated using the engine's specification create bias in favor of compliant implementations	Expand the corpus to include externally annotated ground truth from published benchmarks	Partially mitigated. The 19 externally annotated documents (SemEval 7, MAFALDA 12) provide independent ground truth. Full resolution requires an independent human-annotated gold standard scored blind, identified as highest-priority future work.
Platform: Cross-platform divergence could undermine reproducibility claims	Implement classification logic in Python kernel independent of any LLM; deploy across multiple platforms to validate architecture	Fully materialized, and the materialization validated the mitigation. The $\kappa = 1.000$ on synthetic versus $\kappa = 0.000$ on BLS divergence confirms that LLM-specific processing varies substantially; the deterministic Python kernel routes around the problem.
Platform: LLM provider API changes could require interface-layer updates	Modular interface abstraction layer; minimize per-platform code	Not materialized during the project. Mitigation structure in place for future API changes.
Organizational: Scope creep on an open-ended research project	Explicit proof-of-concept framing; every phase transition documented; future work listed as anticipated artifact rather than failure	Fully mitigated. Scope held to seven specified objectives across six phases. Future work clearly scoped in Section 14.
Organizational: Concentration of implementation responsibility in a single project manager	Code checked into GitHub throughout; automated test suite developed in parallel with core engine; sponsor review at each phase	Managed. 89-assertion pytest suite provides structural safeguard against regression.

10. Issues Encountered

Five categories of issues arose during the project. Each is documented here along with the resolution or current status.

10.1 The Kernel Violation Pattern

The main issue identified in the project was a systematic kernel violation documented in Section 8. Both LLM platforms accurately compute linguistic signal densities at Layer 0 but fail to reliably apply deterministic co-occurrence logic at Layers 2 and 3. This issue was noted across three independent corpora, with the synthetic corpus providing the clearest example: SYN_06 yields a `lex_density` of 0.1220 and `ratio_density` of 0.0488 on GPT, both above the 0.02 threshold for bureaucratic masking, but GPT incorrectly reports it as false. Similarly, SYN_05 produces `ratio_density` and `ethos_density` on Gemini that exceed the thresholds for scientism, yet Gemini also reports it as false. The most significant finding was this compound failure in a single test case.

The solution is not to fix the LLM platforms but to adjust the architecture: the Python kernel should directly execute conditional logic on the signal outputs, regardless of the LLM wrapper's accuracy in reporting flags. For reliable laundering detection in governance applications, LLMs should focus solely on upstream signal estimation, while all Layer 2 and Layer 3 logic must be handled in the kernel.

10.2 SEC Baseline Temporal Shift

The SEC benchmark was revalidated on 2026-03-28. Two Gemini outputs shifted from clean to moral_framing classifications. Document 3.5 (Tupperware/Reuters going-concern article) and document 9.5 (Tesla/Reuters inside-pay-plan article) now fire moral_framing with framing_density of 0.03 exceeding the 0.02 threshold. The gate fires correctly per the kernel specification; the shift is in upstream signal estimation, meaning Gemini's tokenization or density calculation moved between the initial validation (2026-02-15) and the revalidation (2026-03-28).

Table 10-1 documents the two SEC baseline shifts along with the GPT revalidation pattern.

Table 10-1. SEC Baseline Revalidation Shifts

Document	Source	Observation	Classification
Doc 3.5	Tupperware / Reuters editorial	Gemini: clean (initial) to moral_framing (revalidation). framing_density 0.03 exceeds 0.02 threshold.	Genuine Layer 2 firing
Doc 9.5	Tesla / Reuters editorial	Gemini: clean (initial) to moral_framing (revalidation). framing_density 0.03 exceeds 0.02 threshold.	Genuine Layer 2 firing
Doc 8.5	Tesla / Reuters editorial	GPT reports ideological; normative_density 0.0149 does not exceed 0.05; universal_density 0.0025 does not exceed 0.02	Kernel violation KV-07
Doc 3.5, Doc 4.5	Tupperware and Kellanova editorials	GPT reports identity_laundering; ethos_density 0.0188 and 0.0176 do not exceed 0.03	Kernel violations KV-05, KV-06

The methodological implication of the SEC baseline shift is direct: LLM signal estimation is not stable across time. Any publication reporting specific κ or F1 values must specify the revalidation date, and results should be understood as measurements at a point in time rather than permanent properties of the system. The Gemini baseline moved from 20/20 to 18/20 clean on the SEC corpus between validations six weeks apart. A larger shift over a longer time horizon is plausible, as information, time, and the inherent driver of probability of response combinations influence outputs.

10.3 Near-Threshold Classification Sensitivity

Near-threshold classification sensitivity was the most persistent technical issue throughout the project. Speculative financial writing and mixed-register policy documents generate signal densities close to gate thresholds, where small changes in cue-word counts produce classification changes that do not reflect genuine structural differences. The mitigation was to treat the falsifiability flag as one input to the Layer 2 gate hierarchy rather than a standalone classifier, and to flag near-threshold cases in the validation workbook as requiring human review. The underlying issue is a direct consequence of lexicon-based density computation and cannot be eliminated entirely without supervised learning on annotated data, which the project declined to pursue for the reasons documented in Section 4.

10.4 Binary Falsifiability Flag Misclassifications

The binary falsifiability flag produced occasional misclassifications when appropriate scientific hedging language appeared in correctly cautious empirical writing. A research paper that writes 'evidence suggests' in a context where hedging is epistemically appropriate may produce a falsifiability_low value of true under the current cue set. The mitigation was to weight the falsifiability flag as one input to the Layer 2 gate hierarchy rather than a standalone classifier. The ideological gate requires both a density threshold and falsifiability_low; the conjunction reduces false positives on legitimate scientific writing.

10.5 Lexicon Coverage Gaps for Specialized Registers

Cross-platform validation against the SEC benchmark revealed a systematic lexicon coverage gap in initial versions of the engine. SEC-specific vocabulary (million, billion, estimates, material weakness, internal control, obligations, terminated, reimbursement) was not present in the original 424-term lexicon. At the 40 to 86 tokens typical of SEC captured-text summaries, density computation cannot reach the coordinate magnitudes specified in the reference classifications regardless of lexicon coverage. The v2.0 lexicon expansion to 1,091 terms addressed the coverage gap directly, with the largest additions in RATIO (101 terms), HEAD (86 terms), ETHOS (93 terms), and LEX (81 terms). The Kellanova Schedule 13D document (SEC_B05) remains an acute edge case, with a formulaic legal register containing almost no rhetorical vocabulary.

GPT's persistent editorial-register sensitivity on SEC documents (four of six tested editorials fire non-clean gates on GPT while Gemini reads them as clean) is a separate issue from lexicon coverage. This is a classification-logic issue rather than a signal-estimation issue, and it is documented as Kernel Violations KV-05, KV-06, and KV-07 in Appendix D. The v2.1 framing_density threshold recalibration is under investigation as a mitigation, though the underlying issue is that GPT's executed conditional logic departs from the specification even with signals correctly computed.

11. Detailed Project Chronology and Critique

The project was organized into seven phases across twenty weeks. Table 11-1 presents the planned and actual schedule. Phase-level critique follows.

Table 11-1. Project Phase Chronology

Phase	Activity	Planned	Actual
Phase 1	Literature Research and Conceptual Architecture	Weeks 1 to 4	Weeks 1 to 5
Phase 2	Python Kernel Implementation (Layers 0 to 5)	Weeks 5 to 9	Weeks 5 to 10
Phase 3	Validation Workbook Development (v1.0 through v2.1)	Weeks 8 to 12	Weeks 9 to 16
Phase 4	Automated Test Suite Construction	Weeks 11 to 14	Weeks 11 to 14

Phase	Activity	Planned	Actual
Phase 5	Platform Interface Deployments (GPT o3, Gemini, Claude)	Weeks 13 to 18	Weeks 14 to 19
Phase 6	Validation, Analysis, and Documentation	Weeks 17 to 20	Weeks 17 to 20
Phase 7	Revalidation Pass and Kernel Violation Registry	Not planned	2026-03-28

11.1 Phase 1: Literature Research and Conceptual Architecture

Phase 1 involved intensive literature research across four philosophical traditions and the computational NLP literature. The output was the conceptual architecture of the six-layer system and the initial lexicon design across 11 categories.

11.2 Phase 2: Python Kernel Implementation

The Python command-line tool was implemented iteratively, beginning with Layer 0 signal computation. Threshold calibration proved more iterative than anticipated, particularly for Layer 5 coordinate computation, where near-zero signal values produce coordinate placements that do not match visual intuition on clean regulatory text. Each Layer was implemented, tested with hand-coded examples, and then integrated with the previous layers. The phase ran one week over plan. A more systematic threshold optimization approach using pre-coded calibration documents would have made this process more efficient. The initial threshold values were selected based on published

NLP research (LIWC cutoff conventions, General Inquirer density norms) and were not strongly coupled to the engine's specific analytical categories; empirical refinement in later phases adjusted the gate thresholds based on observed false-positive rates.

11.3 Phase 3: Validation Workbook Development

The validation workbook went through four major versions. Version 1.0 included two synthetic seeds and five test documents, while Version 1.1 added twenty SEC benchmark documents. Version 1.2 introduced six synthetic gate-test cases, and Version 1.3 integrated the BLS Validation Corpus with Cohen's κ and F1 scoring. Version 2.0 restructured the corpora by distinguishing between baseline-validation roles and merging analytical dimensions into two primary ones: Classification and Authority Shielding. Version 2.1, completed on 2026-03-28, revalidated the SEC baseline, reduced the BLS corpus from 22 to 12 documents for class-distribution balance, and incorporated MAFALDA External and SemEval Hyperpartisan sub-corpora with externally annotated ground truth.

The phase exceeded the plan by four weeks due to a circularity issue identified during v1.3 development, where reference classifications depended on a comparative review of GPT and Gemini, resulting in a misleading κ score. The restructure in Version 2.0 addressed some circularity, while the v2.1 addition of external annotations resolved the

remainder. The register-mismatch gradient was established thanks to the MAFALDA run, justifying the extended timeline.

11.4 Phase 4: Automated Test Suite

The automated pytest suite validates the deterministic kernel against pre-specified expected outputs for all six layers. 89 test assertions were implemented, all passing in v2.0.0. The test suite executes the Python kernel directly, bypassing the LLM interface. This design choice is why the suite passes 89 of 89 assertions despite the kernel violations documented on LLM platforms: the Python kernel executes the deterministic specification correctly; the LLM wrappers do not always execute it faithfully. The test suite would not catch the kernel violations because the violations occur at the LLM interface boundary, not inside the Python kernel.

11.5 Phase 5: Platform Interface Deployments

The GPT-03, Gemini, and Claude interface implementations were successfully completed in the v2.0 release cycle, resolving all ten cross-platform failure patterns from the previous release. However, the phase extended an extra week due to higher-than-anticipated costs for per-platform system prompt engineering. Each platform requires a prompt that specifies deterministic criteria, defines lexicon, and instructs the system to generate a four-section audit output. Even minor prompt variations can lead to noticeable differences in output quality, though they do not impact kernel-level

classification. A more modular interface could lower costs for future developments, and creating a dedicated application for industry-specific reasoning and linguistic analysis would also be advantageous.

11.6 Phase 6: Validation and Documentation

The final phase comprised complete validation runs across all sub-corpora, validation workbook documentation, and production of this paper along with the companion IEEE conference paper submitted to ICETM 2026. The phase ran on plan. Formal benchmarking against the full MAFALDA dataset (200 annotated texts) and the SemEval-2023 Task 3 corpus is the highest-priority extension and is scoped for the post-project period.

11.7 Phase 7: Revalidation Pass

A post-submission validation pass was conducted on 2026-03-28. This phase was not in the original project plan. It was added after the Phase 6 validation runs produced results that raised two questions. First, whether the Cohen's κ of 1.000 for Gemini on the BLS corpus represented genuine implementation fidelity or a circularity artifact. Second, whether LLM signal estimation was stable across time. The Phase 7 revalidation resolved the first question by adding externally annotated ground truth (SemEval, MAFALDA) to the workbook, and it demonstrated an answer to the second by documenting the SEC

baseline shift. The kernel violation registry documenting 10 confirmed violations was produced during this phase.

11.8 Overall Critique

The project ran six weeks over the original twenty-week plan across all phases combined. The overrun was concentrated in Phase 3 (validation workbook development) and was driven by scope decisions that proved correct in retrospect: expanding from the initial 7-document test set to the 69-document multi-corpus framework, and adding externally annotated ground truth to address the circularity risk. Without those expansions, the two principal findings of the project (kernel violation pattern, register-mismatch gradient) could not have been established empirically. One decision the project would approach differently in retrospect is the initial platform deployment sequence. GPT o3 was deployed first, Gemini second, Claude third. The sequence was convenient but not optimal. A parallel deployment across all three platforms from the start, with identical prompts, would have surfaced the kernel violation pattern earlier in the project and allowed more time for mitigation design. The v2.1 finding that kernel violations appear systematically across platforms is a late-phase discovery that reshaped the project's conclusions. It should have been a mid-phase discovery.

12. Lessons Learned

12.1 Philosophy as Engineering Specification

The most productive lesson of the project was the discovery of how precisely philosophical frameworks can serve as engineering specifications. Popper's falsifiability criterion, once understood at the lexical level, specifies exactly what an unfalsifiable claim looks like. Vague hedging that cannot be pinned down. Universal attributions that no counterexample can touch. Explicit dismissal of contrary evidence. Each of these is realizable as a cue-word count with a threshold. The leap from 'claims structured to be immune from evidence are epistemically inferior' to 'set the evasion gate at evasion_count at least 2 where evasion_count increments on beside the point, no serious person, the committee's findings speak for themselves' is shorter than expected. This suggests a broader research direction: other philosophical frameworks that have resisted operationalization (Habermas on communicative action, Taylor on expressivism, Ricoeur on narrative identity) may be more amenable to deterministic implementation than the literature suggests, provided the operationalization is done at the cue-word level rather than the concept level.

12.2 Determinism as a Feature

The decision to implement the engine deterministically was initially treated as a constraint imposed by the governance use case. The lesson learned is that determinism

is itself a feature. A deterministic tool is fully auditable, fully interrogable, and fully transparent. The kernel violation pattern documented in Section 8 is the strongest empirical confirmation of this lesson: no single LLM implements the deterministic specification consistently across all institutional discourse corpora, which means the deterministic Python kernel is not merely a philosophical preference but a structural necessity for reliable governance applications. The 75 percent classification match rate achieved by the deterministic kernel is, in practice, more useful to decision-makers than a hypothetical 90 percent match rate achieved by a probabilistic classifier that cannot explain its decisions.

12.3 Multi-Platform Performance Is Not Uniform

The most unexpected empirical finding was the performance divergence across platforms. Gemini achieves perfect F1 on SEC and perfect κ on BLS (on the classification dimension) while GPT o3 fires seven moral_framing gates on SEC editorial documents and exhibits majority-class bias on BLS. The lesson is that multi-platform validation should be treated as a primary validation requirement, not a secondary check. Any tool that reports validation results from a single LLM platform is reporting results that reflect that platform's statistical biases in addition to the tool's intrinsic properties. A minimum of two platforms, ideally three or more with different training corpora, is needed to separate platform-specific effects from tool-specific effects.

12.4 Multi-Platform Deployment Complexity

The decision to deploy across three AI platforms added more complexity than anticipated. The model-agnostic Python core successfully isolated the classification logic, but per-platform system prompt engineering was non-trivial. Each platform required a prompt tuned to its specific output formatting conventions, its handling of structured data, and its response to explicit deterministic specifications. The lesson is that multi-platform deployment should be explicitly budgeted as a distinct engineering workstream, with its own time allocation and its own test protocol. Treating platform deployment as a simple wrapper around the core engine underestimates the engineering investment required.

12.5 The Register-Mismatch Principle

The MAFALDA validation finding (κ of 0.053 on the classification dimension) taught a methodological lesson that extends beyond this project. Cross-register transfer of analytical frameworks is not automatic. MAFALDA's *ad hominem* and the engine's evasion gate are both rhetorical-move classifiers, but they operate on different lexical substrates in different communicative contexts. A tool calibrated for institutional register is not effective at auditing informal argumentation. A tool calibrated for informal argumentation is not effective at auditing institutional structural integrity. The analytical boundary is a property of the domain, not a design limitation. Future work

that attempts to unify these domains under a single analytical framework will need to address the register-mismatch gradient explicitly rather than assuming it can be overcome by corpus expansion alone.

12.6 Scope Control in Open-Ended Technical Projects

The explicit proof-of-concept framing was essential to completing a coherent, defensible project within the available time. By defining future work as a legitimate and anticipated project artifact rather than a list of failures, the proof-of-concept scope allowed the project to deliver complete validated outputs against each of the seven core objectives while preserving honest acknowledgment of what was deferred. Section 14 lists the future work explicitly. This is not a list of missing work; it is a list of work scoped out of the proof of concept and identified as next steps. The distinction matters for project defense and for sponsor communication.

12.7 The Value of Revalidation

Phase 7 was unplanned and arose from questions raised by Phase 6 results. The revalidation pass produced the two most consequential findings of the project: the kernel violation registry and the SEC baseline temporal shift. Without Phase 7, neither would have been documented, and the paper would have presented results that were technically correct but incomplete. The lesson is that the standard project lifecycle (plan, execute, close) is insufficient for research that depends on LLM-based tooling,

where the tooling itself is non-stationary. A revalidation phase should be planned into the project from the start, not added retrospectively.

13. Conclusions and Summary

13.1 Research Question 1: Can the System Be Built?

The answer is yes, with qualifications. The six-layer deterministic architecture was implemented in Python with full test suite coverage (89 assertions, all passing) and operates correctly and reproducibly across all 69 tested documents and all three deployment platforms (GPT o3, Gemini, Claude). The 100 percent match rate on all six synthetic gate-test cases confirms deterministic kernel correctness at the unit level across every classification pathway.

The qualification is the kernel violation pattern. Both LLM platforms correctly compute the 14 Layer 0 signal densities but do not always execute the Layer 2 and Layer 3 conditional logic faithfully. Ten confirmed violations were documented across three independent corpora during the 2026-03-28 revalidation. The implication is clear: the system can be built deterministically, but the deterministic behavior is reliable only when the conditional logic is isolated in the Python kernel rather than delegated to the LLM wrapper. The architecture accommodates this: the LLM handles signal estimation and audit-report formatting, the Python kernel handles all classification logic, and the boundary between the two is documented and auditable.

The BLS and Energy validation corpora provide empirical evidence of RQ1 completion. Gemini achieves κ of 1.000 and F1 of 1.000 on the BLS Validation Corpus classification dimension. On the Energy corpus, Gemini achieves the same κ of 1.000 against the reference classifications while GPT o3 achieves κ of 0.302 (Fair). Both models converge on moral_framing detection for ENE_X5E, the NPR/AP/CNN reconstruction of the 2023 OPEC+ surprise cuts, providing the strongest single-document cross-platform convergence in the Energy corpus.

13.2 Research Question 2: Are the Outputs Useful?

The answer is yes. The BLS delta analysis with its measured shift from clean, zero intensity, open-loop reasoning, and no laundering to evasion, Q4 quadrant, intensity near 6.2, defensive routine, and identity laundering is the clearest single demonstration of practical utility. It provides a quantified reproducible characterization of what the White House statement does to the BLS data structurally: it transforms a neutral statistical correction into a politically immunized conclusion.

The SEC benchmark corpus demonstrates practical utility for financial disclosure analysis. All ten technical filings exhibit bureaucratic masking with defensive-routine posture in the HEAD+STABILITY quadrant. All ten Reuters editorial interpretations classify as open-loop reasoning with no laundering in the HEAD+DISRUPTION quadrant. Given two documents covering identical corporate events, the engine reliably

differentiates the filing's institutionally anchored stability-coded register from the editorial's market-focused disruption-coded interpretation.

The Energy benchmark corpus extends this utility to commodity-price journalism.

Primary WSJ and FT articles on the 2020 negative WTI price crash and the 2023 Saudi unilateral cut do not trigger laundering flags despite high disruption density, because their quantitative claims are directly sourced and analytically grounded. Syndicated wire reconstructions of the same events trigger scientism, the compound of quantitative data with institutional authority cues delegated to external sources. The engine distinguishes source citation (analytically integrated) from authority laundering (delegated epistemic weight) via the density co-occurrence threshold.

13.3 Primary Contribution

The primary contribution of the project is a working proof of concept demonstrating that the philosophical concepts of falsifiability, organizational learning, language-game analysis, and epistemic humility can be operationalized as a deterministic computational system that produces practically useful structural audits of institutional text. The 69-text validation framework, the 19 externally annotated documents, and the 10-violation kernel registry collectively establish the capabilities and the limits of this approach.

13.4 Three Principal Findings

Three findings emerged from the revalidated data (2026-03-28).

First, the kernel violation pattern. Contemporary LLM platforms can correctly compute the linguistic signal densities required for structural auditing but cannot be relied upon to execute the deterministic conditional logic that converts those signals into classification and laundering flags. This finding is consequential beyond the Epistemic Engine. It suggests that any governance tool that delegates conditional logic to an LLM interface layer will exhibit similar systematic violations, and that isolating the conditional logic in a deterministic kernel is the correct architectural response.

Second, the register-mismatch gradient. The engine's analytical categories produce high-fidelity classification on institutional text (κ of 1.000 on synthetic institutional, F1 of 1.000 on Gemini SEC) and near-zero classification on informal argumentation (κ of 0.053 on MAFALDA). This confirms that institutional structural constructs (evasion, authority laundering, defensive routines) are domain-specific to institutional registers and do not map onto general-purpose argumentation constructs, establishing that structural epistemic auditing and general fallacy detection are distinct analytical tasks.

Third, the scale contrast with institutional annotation infrastructure. The JRC campaigns required over seventy scholars and months of calibration to annotate 2,500 documents with persistent expert disagreement. The deterministic kernel produces reproducible output on 69 documents with fully documented failure modes and no inter-annotator calibration requirement. The cost of this determinism is specificity: the kernel's thresholds are fixed, its lexicons are domain-bounded, and its failure modes

(documented in the violation registry) are transparent. The JRC campaigns' strength is breadth, multilingual coverage, and the capacity to capture constructs that no fixed threshold can express. Neither approach dominates the other. For the specific governance use case of screening institutional text for structural epistemic properties at scale with reproducible outputs, the deterministic kernel offers zero marginal cost per document and zero inter-rater variability.

14. Limitations, Recommendations, and Scope for Future Work

14.1 Current Limitations

The engine's primary limitation is the gap between lexicon-based signal estimation and the richer contextual understanding required for fully general structural auditing. The MAFALDA result quantifies this boundary: on informal argumentative text outside the engine's institutional calibration domain, signal densities are insufficient to trigger gate and laundering classifications. Seven of twelve MAFALDA texts are short informal sentences (5 to 25 words) that produce near-zero signal densities on both platforms, explaining the κ gradient from 1.000 on synthetic institutional text to 0.053 on MAFALDA informal argumentation.

The evasion and ideological gates remain empirically thin outside synthetic texts. One confirmed ideological case (BLS_B06E, the PIIE Kolko piece) and one confirmed evasion case (SemEval Article 1) exist in the validation corpora, with zero confirmed non-

synthetic cases on MAFALDA. Expanding the corpus with documents whose institutional registers are more likely to activate these gates (ESG corporate sustainability reports, political press statements responding to economic data releases, and discretionary earnings call transcripts) would provide the non-clean distribution needed to compute per-class precision and recall across all four gate outcomes.

The SEC baseline temporal shift (Gemini 18/20 clean on revalidation, from 20/20 on initial validation six weeks earlier) demonstrates that LLM signal estimation is not stable across time. Any publication citing specific κ or F1 values must specify the revalidation date, and results should be understood as measurements at a point in time rather than permanent properties of the system. The shift has direct implications for longitudinal governance deployments: if the engine is deployed in a compliance workflow, the signal-estimation step must be monitored for drift, and recalibration protocols must be established.

The Layer 3 kernel violations are systematic enough to raise the question of whether laundering detection is reliably deployable without the Python kernel enforcing the co-occurrence logic independently of the LLM. The revalidation data supports a clear recommendation: for governance applications requiring reliable laundering detection, the Python kernel must execute the conditional logic; the LLM should be restricted to upstream signal estimation and audit-report formatting.

Cross-linguistic validity is untested. All tested documents are English-language U.S. institutional texts. The engine's lexicons were developed on U.S. regulatory, labor-market, and financial-journalism registers. Non-U.S. English institutional discourse (UK regulatory, Canadian policy, Australian corporate) is expected to require lexicon extension; non-English registers require both lexicon development and tokenization adjustment.

14.2 Recommendations for Institutional Deployment

For organizations considering deployment of the engine, four recommendations follow from the validation data.

First, use the engine as a triage and prioritization tool rather than as a final classifier.

The 75 percent overall match rate means approximately one in four documents may require human review. The engine's strongest use is flagging the documents that merit closer expert attention, not producing final compliance determinations.

Second, extend the lexicons for domain-specific deployment. Organizations deploying the engine in non-U.S. contexts, specialized technical domains, or industry-specific registers should invest in domain-specific lexicon extension following the v2.0 function-first methodology documented in the project repository.

Third, deploy the delta analysis capability for high-value paired document comparisons.

The engine's strongest validated use case is the structural comparison of paired

documents covering identical facts. The BLS release / White House statement delta, the SEC filing / Reuters editorial delta, and the WSJ primary / wire reconstruction delta are all validated use cases. Single-document analysis is supported but less discriminating. Fourth, apply the Inter-Model Agreement data as a platform-selection guide. For SEC-type institutional filings, Gemini's outputs are more reliable on the Authority Shielding dimension. For BLS-type economic reporting, Gemini achieves κ of 1.000 on classification while GPT o3 exhibits majority-class bias. The v2.0 Python kernel addresses both platforms, but the recommendation is to run the deterministic kernel directly rather than rely on the LLM wrapper to execute classification logic.

14.3 Scope for Future Work

The most impactful near-term improvements are organized into seven items.

Completing and validating the v2.1 SEC lexicon expansion, including framing_density threshold recalibration to address the GPT editorial-register gate activations documented in the kernel violation registry. This work is under way and is scoped for the post-project period.

Constructing an independent human-annotated reference standard scored blind. The SEC-anchored reference classifications in the validation workbook were derived from a comparative review adjudicating GPT and Gemini divergences against the engine's specification. Even with the external MAFALDA and SemEval ground truth, a blind-

scored human reference on the institutional corpora would strengthen claims beyond implementation fidelity to independent accuracy.

Expanding the corpus with ESG corporate sustainability reports, political press statements, and discretionary earnings call transcripts to increase non-clean gate distribution across the evasion and ideological classes. These registers are expected to exhibit the structural patterns the engine is designed to detect, and their absence from the current corpus is the single largest gap in the validation distribution.

Developing a supervised signal-estimation layer trained on reference-classification cases with MAFALDA-coded annotations as labels. This would augment the current lexicon-based signal estimation with learned representations while preserving the deterministic Layer 2 and Layer 3 logic. The architectural separation between Layer 0 and Layers 2 through 5 supports this extension without requiring kernel changes.

Formal benchmarking against the full MAFALDA dataset (200 annotated texts) and the SemEval-2023 Task 3 corpus. The current engagement uses 12 MAFALDA texts and 7 SemEval articles; scaling to the full datasets would allow precision-recall analysis across the complete fallacy and persuasion-technique taxonomies.

Extending the laundering taxonomy from the current three flags (scientism, bureaucratic masking, identity laundering) toward SemEval-2023 Task 3's 23-technique set. The extension is a designated near-term priority. A fourth flag for distance

laundering, the deliberate emotional and cognitive distancing of readers from the structural implications of factual data, is in design and is candidate for the v2.2 release. Using the Inter-Model Agreement data as a longitudinal tracking instrument to measure κ and F1 improvement across future engine versions. The Inter_Model_Agreement sheet in the v2.1 workbook provides the baseline; subsequent revalidation passes can use the same scoring structure to quantify version-over-version improvement, which is a design pattern the LLM governance community currently lacks.

15. Literature Survey

15.1 Introduction to the Literature

The Epistemic Engine is situated at the intersection of six bodies of literature that jointly inform its design, evaluation, and theoretical framing. These literatures include the philosophical traditions that shape its evaluative categories, critical discourse analysis, computational natural language processing, fallacy and persuasion benchmarks such as MAFALDA and SemEval, institutional annotation frameworks for values and persuasion detection, and scholarship on the epistemic risks of large language models and human-AI convergence. Together, they provide the conceptual and technical basis for analyzing how institutional discourse encodes authority, persuasion, and epistemic risk.

15.2 Philosophical Foundations

15.2.1 Kierkegaard: Epistemic Humility as a Design Principle

The engine's design is influenced by Kierkegaard's view of truth as subjectivity (Kierkegaard, 2009). In the Concluding Unscientific Postscript, he rejects the idea of an absolute objective standpoint for evaluating human discourse, emphasizing that individuals are always situated within their own perspectives. This perspective informs the engine's choice to maintain interpretive uncertainty instead of producing definitive judgments. The system is deterministic and reproducible with the same inputs, and it does not attempt to resolve differing worldviews. Features like the continuous-valued coordinate spectrograph and the preservation of inter-model disagreement as data exemplify this approach.

15.2.2 Wittgenstein: Language Games and Institutional Meaning

The engine's Layer 1 classifier derives from Wittgenstein's concept of language games (Wittgenstein, 2009). Wittgenstein argued that language derives its meaning not from abstract correspondence to reality but from its use within specific social practices, what he called forms of life. Different communities, institutions, and disciplinary traditions constitute different language games, each with its own internal rules for what counts as a valid move.

Deitz and Arrington apply Wittgenstein's framework to behavioral and cognitive science, demonstrating that the language-game concept allows practitioners to evaluate the meaning of terms not by reference to internal mental states but by their patterns of use within discourse communities. This application is directly relevant to the Epistemic

Engine, which treats all text as institutional discourse with identifiable rules, conventions, and rhetorical strategies. Ellaway (2021) extends this analysis to scholarly writing in health professional education, observing that scholarly writing conventions can both enable and constrain inquiry. This observation maps directly onto the engine's concern with authority laundering: the use of academic, bureaucratic, or quantitative form to foreclose rather than invite contestation.

The Wittgensteinian framework carries a methodological consequence that the project's MAFALDA validation confirmed empirically: the analytical boundary between institutional structural auditing and general fallacy detection is a property of the domain, not a design limitation. The register-mismatch gradient from κ of 1.000 on synthetic institutional text to κ of 0.053 on MAFALDA informal argumentation is the measurement of that boundary.

15.2.3 Popper: Falsifiability and the Bias Gate

The engine's Layer 2 bias gate implements Popper's falsifiability criterion directly (Popper, 2002). For Popper, the demarcation between science and non-science turns on whether claims are structured in a way that allows them to be proven false by evidence. Claims that are systematically insulated from refutation (through vagueness, universality, or what Popper called immunizing stratagems) forfeit their claim to epistemic authority while often amplifying their rhetorical force. The engine operationalizes this insight through its gate hierarchy, which evaluates four sequential

conditions: whether the text exhibits evasion, ideological structure, moral framing, or passes all gates as structurally clean. The output uses Popper's vocabulary directly and applies it as an automated structural test.

Bhardwaj (2025) reinforces the Popperian concern by showing how theorists employ rhetorical devices (metaphor, analogy, and appeals to value consensus) to structure arguments in ways that make them difficult to contest. This finding underscores the need for an automated tool capable of distinguishing legitimate argument structure from rhetorical self-protection. The MAFALDA validation later confirmed that Popper's construct of immunizing stratagems is register-specific to institutional discourse: the vocabulary of institutional evasion differs from the vocabulary of informal argumentative evasion, and a single tool cannot detect both at the same time.

15.2.4 Argyris: Single-Loop and Double-Loop Learning

The engine's Layer 4 posture classifier derives from Argyris's theory of organizational learning (Argyris, 1976). Argyris introduced the distinction between single-loop and double-loop learning. In single-loop learning, organizations respond to errors by adjusting actions within existing governing norms; the system corrects itself without examining the assumptions that produced the error. In double-loop learning, the governing norms themselves are subjected to scrutiny and revision.

Argyris further introduced the concept of defensive routines: organizational behaviors that protect individuals and groups from the discomfort of examining their own

assumptions. The engine's Layer 4 tests whether discourse manifests defensive routines by examining whether Layer 3 authority laundering is present, treating authority laundering as a structural proxy for the kind of discourse that protects governing norms from examination. Wright (2024) provides a contemporary application of Argyris's framework to management research, arguing that advocacy-based scholarship functions as a form of single-loop learning at the disciplinary level, producing arguments that confirm governing value commitments rather than subjecting those commitments to scrutiny.

15.2.5 Critical Discourse Analysis

Fairclough (2015) established that the register, claim structures, and deployment of authority in institutional language are never neutral. Van Dijk (1998) demonstrated that discourse is a primary site where social power is exercised and contested. Entman (1993) showed that selecting and emphasizing certain aspects of a perceived reality shapes how audiences understand causation, moral evaluation, and appropriate response. Berger and Luckmann (1966) established that institutional language is a primary mechanism by which socially constructed realities are made to appear objective and given. This tradition provides the bridge between the philosophical foundations described above and the computational methods described below: it theorized for decades that structural properties of discourse can be identified and analyzed, and it is the lineage the Epistemic Engine computationally operationalizes.

15.3 Natural Language Processing and Text Analysis

15.3.1 Word Embeddings and Semantic Resources

The computational backbone of the engine's raw-text mode is lexicon-based density computation. This approach draws on the tradition of distributional semantics in NLP, in which the meaning of words is represented by their patterns of co-occurrence in large corpora. Rothe and Schütze (2017) developed AutoExtend, a system that extends word embeddings to incorporate semantic resources such as WordNet, GermaNet, and Freebase. By learning embeddings for non-word objects such as synsets and lexical entities, AutoExtend enables richer representations of meaning that capture not only distributional similarity but also structured semantic relationships. For the engine, this line of research informs the cue-word lexicons that drive signal measurement, requiring lexicons that capture semantic clusters rather than surface-form matches alone.

15.3.2 Lexicon-Based Density Computation

The engine's signal-estimation approach extends a computational tradition originating with the General Inquirer (Stone, Dunphy, Smith, and Ogilvie, 1966) and the Linguistic Inquiry and Word Count system, LIWC (Pennebaker, Francis, and Booth, 2001). The General Inquirer demonstrated that counting the frequency of words belonging to predefined semantic categories could reliably capture structural properties of discourse including evaluative tone, cognitive complexity, and institutional register. LIWC refined

this approach into a widely validated instrument for measuring psychological and linguistic dimensions of text through dictionary-based word counting. The engine's Layer 0 operates in this tradition, computing 14 signal densities via lexicon-based cue counting normalized per token. The tradition from Stone through Pennebaker through Zadeh provides the theoretical warrant for treating word-frequency ratios as reliable indicators of structural properties.

15.3.3 Computing with Words

Zadeh's (2012) Computing with Words framework addresses a fundamental limitation of classical symbolic computation: natural language is inherently imprecise, and much of what humans communicate is conveyed through words whose meaning is context-sensitive, gradable, and resistant to binary classification. Zadeh proposes a computational architecture based on fuzzy logic and fuzzy set theory in which words are treated as granular labels for fuzzy sets, enabling inference over linguistic rather than numerical representations. The engine's coordinate spectrograph places text on continuous HEAD/HEART and DISRUPTION/STABILITY axes rather than forcing discrete classification, reflecting Zadeh's insight that natural-language categories are inherently gradable.

15.3.4 Detecting Harmful and Biased Discourse

Fortuna and Nunes (2019) surveyed the state of automatic hate speech detection, documenting the range of algorithms, feature representations, and annotated datasets

developed to identify harmful language online. Their survey reveals several features of the detection problem relevant to epistemic auditing. Hate speech detection requires contextual understanding beyond keyword matching. The target concept is a normative construct whose boundaries vary across legal systems, platforms, and communities. Annotation inconsistency is a persistent challenge. Each of these findings reinforces design choices in the engine: it deliberately avoids classifying discourse as hateful or harmful, reporting structural properties that are closer to the surface form of the text and thus more amenable to reliable automated detection.

Pan and Yang (2025) extend the practical stakes from social media to corporate disclosure. Their finding that the structural quality of corporate information disclosure functions as a signaling mechanism that reduces information asymmetry and materially influences investment efficiency establishes that how institutional text communicates (not merely what it reports) produces measurable downstream consequences. This finding provides direct empirical support for the project's use-case framing: the SEC benchmark corpus is not an abstract test set; the structural properties the engine measures on that corpus are known to correlate with investment outcomes.

15.4 Fallacy Detection, Argumentation, and NLP Benchmarks

15.4.1 MAFALDA: Taxonomy, Benchmark, and the Register-Mismatch Gradient

The MAFALDA benchmark (Helwe et al., 2024) provides a comprehensive current evaluation framework for automated fallacy detection. MAFALDA merges and unifies previous fallacy datasets under a refined taxonomy of 12 fallacy types across three top-level categories (Pathos, Ethos, and Logos) with manual annotations across 200 texts and a disjunctive annotation scheme that accommodates inherent subjectivity. The best inter-annotator agreement among MAFALDA's expert annotators was 0.4 at Level 1, a level the authors characterized as very low, confirming the innate difficulty of fallacy detection even for trained human raters.

MAFALDA's evaluation of several language models under zero-shot settings reveals a consistent finding: fallacy detection is substantially more difficult than it might appear, because fallacious arguments often conform to the surface structure of valid reasoning while failing at the level of inferential warrant. The strongest performer (GPT-3.5) showed significantly degraded accuracy at the most granular classification level, precisely where the distinction between vocabulary presence and rhetorical function matters most.

Twelve MAFALDA texts covering the taxonomy's three top-level categories were processed blind by both engine implementations. Both platforms systematically underfired against the expected gate labels: Gemini produced κ of 0.053 (slight agreement) and GPT produced κ of 0.000. The result is driven by register mismatch. Seven of twelve texts are short informal sentences (5 to 25 words) that produce near-

zero signal densities on both platforms. The evasion gate requires `evasion_count` of at least 2; informal personal attacks produce `evasion_count` of 0. The institutional evasion the engine detects operates through multi-sentence hedging, deflection, and contest-blocking vocabulary that short informal text does not contain. This register mismatch runs in both directions: a tool calibrated for informal argumentation fallacies is not effective at auditing the structural integrity of a regulatory filing, and a tool calibrated for institutional structural auditing is not effective at detecting ad hominem attacks in informal contexts. The MAFALDA run proves this empirically rather than asserting it. The strongest single-document finding is `MAF_02`, a textbook appeal to false authority ('Four out of five dentists recommend Happy Glossy toothpaste. Therefore, it must be great.'). Gemini detects scientism on Layer 3, correctly identifying the compound of statistical quantification ('four out of five') with credential invocation ('dentists'). This is the structural property the engine was designed to surface, correctly identified on Layer 3 (authority laundering) rather than Layer 2 (bias gate), exactly as the architecture predicts for Ethos fallacies.

15.4.2 SemEval Evaluation Tasks

SemEval-2023 Task 3 (Piskorski et al., 2023) addresses framing and persuasion technique detection across nine languages using a taxonomy of 23 persuasion techniques organized across three tasks: genre classification, framing detection, and persuasion technique identification. The engine's Layer 1 language-game classifier maps

directly to SemEval-2023 Task 3's genre classification. The 23-technique persuasion taxonomy provides an explicit roadmap for extending the engine's three-type laundering detection to a richer set; this extension is a designated near-term priority. SemEval-2020 Task 11 (Da San Martino et al., 2020) demonstrated that technique classification proved far harder than span identification, with under-sampled categories systematically misclassified regardless of model architecture. This finding has direct implications for the engine's own class-distribution limitation: the evasion and ideological gates remain empirically thin outside synthetic texts.

The project's validation framework includes seven externally annotated articles from the SemEval hyperpartisan corpus, processed by both engine implementations and scored against independent ground truth produced entirely without engine involvement. Inter-model agreement on the SemEval corpus of $\kappa = 0.462$ (moderate) falls between synthetic institutional text ($\kappa = 1.000$, perfect) and MAFALDA informal argumentation ($\kappa = 0.053$, slight). This gradient is itself informative: agreement degrades as the input register moves away from the engine's institutional calibration domain.

15.5 Institutional Annotation Frameworks: Values and Persuasion

Detection

Two technical reports produced by the European Commission's Joint Research Centre provide important methodological precedent and empirical grounding for the Epistemic Engine's design. The persuasion report (Piskorski et al., 2023a) examines persuasion technique identification across six languages with more than thirty annotators. The values report (Reitis-Münstermann et al., 2024) examines values detection across nine languages with approximately seventy scholars. Read together, these reports suggest a structural relationship between values and persuasion as complementary influence systems in text, while also showing that no existing tool addresses both simultaneously.

15.5.1 The Scale and Institutional Scaffolding Problem

Both JRC campaigns begin from the premise that text production has outpaced human analytical capacity. The persuasion guidelines operationalize this across roughly 1,000 documents in six languages with over thirty annotators. The values campaign deployed approximately seventy scholars across nine languages. The institutional infrastructure required to produce these annotations was substantial: training materials, weekly calibration meetings, multi-phase curation processes, language-group leads, and a central authority resolving disagreements. Together, the two campaigns annotated over 2,500 documents with months of institutional investment in calibration.

15.5.2 The Expert Disagreement Problem

Perhaps the most consequential finding across both campaigns is the persistence of expert disagreement even under controlled conditions. The values guidelines

acknowledge that complete agreement among annotators is unrealistic and instruct annotators not to expend excessive effort pursuing it. The persuasion guidelines report that annotators with prior fact-checking experience systematically conflated persuasion detection with claim verification, and that annotators tended to over-rely on technique names as their sole guidance, leading to misapplication of labels. If trained domain experts operating within elaborate quality-assurance systems still produce inconsistent readings, untrained decision-makers operating under time pressure are substantially more vulnerable to the influence patterns these frameworks document.

15.5.3 The Fact-Check Fallacy

Both JRC frameworks converge on a finding that is critical for the Epistemic Engine's positioning: factual accuracy is orthogonal to the influence architecture of a text. The persuasion guidelines contain an emphatic warning that persuasion technique detection should not be confused with fact-checking, because a factually true text can still deploy persuasion techniques. The values guidelines make a parallel move, instructing annotators to ignore whether information is true or false and label only the value expressed as written. This convergence directly validates the engine's core design principle: a factually accurate briefing can still deploy authority laundering, moral framing, and evasion while encoding defensive routines. Decision-makers who equate verified facts with informed decisions are exposed to precisely the influence dimensions that fact-checking misses.

15.5.4 The JRC Scale Contrast

Both JRC campaigns operated across multiple languages and required substantial institutional investment. The scale contrast between the JRC approach and the Epistemic Engine is itself a finding. Where the JRC campaigns required seventy scholars and months of calibration to annotate 2,500 documents with persistent residual disagreement, the deterministic kernel produces reproducible output on 69 documents with fully documented failure modes and no inter-annotator calibration requirement. The cost of this determinism is specificity: the kernel's thresholds are fixed, its lexicons are domain-bounded, and its failure modes are transparent. The JRC campaigns' strength is breadth: multilingual coverage, diverse annotator perspectives, and the capacity to capture constructs that no fixed threshold can express. Neither approach dominates the other. For the specific governance use case of screening institutional text for structural epistemic properties at scale, the deterministic kernel offers a capability the annotation-campaign model cannot: zero marginal cost per document and zero inter-rater variability.

15.6 Artificial Intelligence, Truth, and Epistemic Risk

15.6.1 The Stochastic Parrot Problem

Bender, Gebru, McMillan-Major, and Shmitchell (2021) offer a foundational critique of large language models directly relevant to the engine's design rationale. The paper

argues that LLMs trained on massive text corpora are, in an important sense, stochastic parrots: they produce linguistically coherent output by modeling statistical regularities in training data without any grounded understanding of meaning, reference, or truth. Because the training data reflects the distribution of text produced by humans (including its biases, fallacies, and ideological commitments) LLMs are liable to reproduce and amplify these patterns at scale.

The project's validation data provides concrete empirical evidence for the Bender et al. concern. Both LLM platforms correctly compute linguistic signal densities at the input layer but fail to execute the deterministic co-occurrence logic required to flag authority laundering at Layer 3. This kernel violation pattern, documented across three independent corpora, demonstrates a specific form of the stochastic parrot problem: the platforms can perform statistical pattern matching on text (signal estimation works) but cannot reliably follow a deterministic specification (conditional logic execution fails). This disconnect has a direct implication for governance applications: a deterministic specification does not guarantee deterministic execution when the upstream signal estimation is delegated to a probabilistic model.

15.6.2 The Human-AI Convergence

Ng (2026) examines the broader societal implications of what he calls the Human-AI Convergence, the progressive integration of artificial intelligence into cognitive, social, and institutional processes. The Hybrid Mind argues that this convergence is reshaping

how knowledge is produced, validated, and transmitted. As AI systems become increasingly embedded in epistemic practices (drafting reports, generating summaries, producing policy analyses, synthesizing research) the structural properties of AI-generated discourse become a matter of institutional concern. An AI-generated policy document that employs immunizing stratagems and defensive routines is epistemically problematic for the same reasons as a human-authored one.

15.6.3 Digital Shadows and Epistemic Accountability

Bhardwaj (2026) extends Plato's allegory of the cave to the contemporary condition of digital media and AI-generated content. The paper argues that digital representations, whether data visualizations, AI summaries, or algorithmically curated feeds, function as shadows on the cave wall: they present themselves as faithful representations of reality while being artifacts of specific computational, economic, and ideological processes. Viewers who lack the means to step outside the cave and examine the light source are liable to mistake the shadow for the substance. By surfacing the structural properties of discourse, the engine's audit function allows users to distinguish between genuine argumentative openness and rhetorical closure dressed in the form of authority.

15.7 Synthesis and Implications for the Applied Project

Four points emerge from this survey. The engine is philosophically grounded in major traditions of scholarship (Kierkegaard, Wittgenstein, Popper, Argyris) with a critical

discourse analysis lineage for treating discourse structure as measurable and consequential. MAFALDA results suggest the engine is domain-specific to institutional registers, as shown by the register-mismatch gradient across synthetic, news, and informal texts.

The engine is computationally feasible. The NLP methods reviewed here provide a solid technical basis, while known limitations such as context sensitivity and annotation variability are addressed through conservative heuristics, layered deterministic design, and explicit failure modes. The kernel violation finding clarifies the boundary between LLM-based signal estimation and logic that must be enforced in Python.

The engine is institutionally validated by precedent. The JRC annotation campaigns show that identifying values, persuasion, and framing in institutional text requires structured, multi-layered analysis and still produces expert disagreement, which the engine avoids through deterministic execution at the cost of domain specificity.

The engine is timely and needed. As AI-generated discourse expands, structural auditing of institutional text has become a practical necessity. Across 69 texts and 19 with external ground truth, the evidence supports three core findings: kernel execution limits, register-specific construct behavior, and a cost-reliability contrast with manual annotation.

16. Situational Analysis

This section incorporates the Situational Analysis prepared earlier in the project, updated with the v2.1 revalidation findings.

16.1 Environmental Analysis

The Epistemic Engine operates at the intersection of five macro trends. First, accelerating LLM integration in institutional communication. Corporations, government agencies, healthcare systems, and academic institutions are increasingly producing and consuming text generated or polished by AI tools. Fluency has improved dramatically; epistemic quality has not. AI-generated institutional communications routinely exhibit the same patterns the project is designed to detect: agency erasure, abstraction from verifiable claims, and strategic evasion that sounds substantive while committing to nothing.

Second, the intensification of institutional discourse as a site of political contestation.

The September 2025 BLS preliminary benchmark revision and the White House statement on the same data are a representative case: the same factual release generated categorically different structural postures from different institutional authors.

The proliferation of this pattern (technical release plus discretionary reinterpretation) motivates the paired-document comparison use case that the engine's delta analysis capability supports.

Third, the development of AI governance regulation. The European Union's AI Act, adopted in 2024, mandates transparency, explainability, and accountability for AI systems deployed in high-risk contexts. The NIST AI Risk Management Framework and emerging ISO standards similarly identify epistemic transparency as a core governance requirement. These regulatory frameworks create direct institutional demand for tools with deterministic, interrogable outputs, which is precisely the engine's design profile.

Fourth, the ESG reporting accountability movement. Environmental, Social, and Governance reporting has become a dominant form of institutional communication. As pressure from investors, regulators, and the public has grown, organizations have developed sophisticated linguistic strategies for communicating accountability without being accountable. Terms such as non-conformance event, workforce optimization, and implementation challenges are systematic strategies for managing perception while obscuring agency and causality. The engine's laundering detection layer is designed to surface these patterns automatically.

Fifth, the emergence of large-scale publicly funded annotation infrastructure. The JRC persuasion and values campaigns operationalized multi-dimensional frameworks for the analytical dimensions the engine targets and confirmed that these dimensions are consequential at institutional scale. The scale contrast between those campaigns and the engine's deterministic kernel is discussed in the project's principal findings.

Across computational linguistics more broadly, annotated datasets sufficient to enable reliable automated structural auditing of institutional discourse do not yet exist. MAFALDA's F1 of 0.138 on fine-grained fallacy detection, SemEval-2023 Task 3's limited institutional-discourse coverage, and the JRC campaigns' documented expert disagreement across months of annotation effort converge on this point. The engine's validation workbook (v2.1, 69 texts with 19 externally annotated ground-truth documents) is the field's most complete annotated resource specifically targeting structural epistemic auditing of institutional discourse. Expanding that resource is both the engine's primary development priority and its primary research contribution.

16.2 Competitive Landscape

The engine occupies a distinctive position with no current commercial product offering deterministic multi-layer structural auditing of institutional text with explicit philosophical grounding and model-agnostic architecture.

Commercial text analysis tools, including IBM Watson Natural Language Understanding, AWS Comprehend, Azure Text Analytics, and Amazon Comprehend, offer sentiment analysis, topic modeling, and named entity recognition. None performs structural epistemic auditing. Fact-checking tools including ClaimBuster operate at the claim level and cannot detect structural immunizing stratagems. Financial NLP tools including FinBERT perform sentiment classification on financial text and were not trained on data

designed to capture structural epistemic properties. Readability metrics measure surface complexity, orthogonal to epistemic properties. Toxicity detectors are similarly orthogonal; institutional language designed to obscure accountability is almost never toxic.

Academic benchmarks including MAFALDA and SemEval-2023 Task 3 provide taxonomy references and evaluation frameworks but are not tools. MAFALDA's 200 annotated texts establish the performance ceiling achievable with the largest available fallacy corpus. SemEval's persuasion taxonomy provides a roadmap for extending the engine's laundering set. Neither produces integrated multi-layer structural audits in deterministic form.

The JRC campaigns produced annotation infrastructure rather than tools. Their relevance is methodological and empirical. They validated the problem domain at institutional scale, confirmed that factual accuracy is orthogonal to influence architecture, and documented the human cost of producing domain-specific annotated data at scale. None of these institutional frameworks produces automated structural audits.

16.3 Stakeholder Analysis

The primary stakeholders are governance and compliance professionals screening institutional communications for disclosure risk. For this group, the engine provides

reproducible structural audits at zero marginal cost per document and zero inter-rater variability. The SEC benchmark corpus results (nine of ten technical filings exhibit bureaucratic masking with defensive-routine posture; all ten paired editorial interpretations classify as open-loop reasoning without laundering) demonstrate the delta-analysis capability for financial disclosure screening.

Investment analysts and financial research professionals constitute a second primary stakeholder group, for whom the structural delta between SEC regulatory filings and discretionary executive communications is a high-stakes analytical problem. No existing NLP tool, including FinBERT, was trained on data designed to capture this structural delta at the register level. The SEC and Energy benchmarks empirically demonstrate it across 30 real-world documents.

Policy analysis and journalism organizations need structural audits of institutional rhetoric rather than sentiment scoring or fact-checking alone. The BLS delta analysis (clean / zero intensity versus evasion / Q4 / 6.2 intensity on identical underlying data) is the clearest single demonstration of the utility these organizations currently lack.

NLP and computational linguistics researchers form a fourth stakeholder group. For them, the engine's contribution extends beyond the tool itself to the research infrastructure: the 69-text validation workbook with MAFALDA-aligned annotations, the inter-rater reliability data (Cohen's κ and F1 across two merged dimensions on six

validation corpora), the externally annotated ground truth from SemEval and MAFALDA, and the kernel violation registry documenting systematic LLM execution gaps.

Academic researchers in management, organizational communication, and critical discourse analysis form a fifth stakeholder group. For them, the engine demonstrates that philosophical and organizational-theory frameworks (Popper on falsifiability, Argyris on defensive routines, Wittgenstein on language games) can be operationalized computationally without sacrificing conceptual integrity. The translation from philosophical concept to deterministic classification logic is documented and reusable.

16.4 Porter's Five Forces Analysis

Table 16-1 presents the Porter's Five Forces assessment for the Epistemic Engine, updated to reflect the v2.1 validation findings.

Table 16-1. Porter's Five Forces Assessment

Force	Intensity	Analysis
Threat of New Entrants	Moderate	Barriers have dropped with open-source LLMs, but the theoretical framework requires domain expertise not trivially replicated. The JRC campaigns demonstrate that even well-resourced institutional teams require months of calibration to produce consistent annotations. ML-based entrants must first solve an annotated-corpus problem for which no published resource exists at sufficient scale. The deterministic kernel sidesteps this burden.
Bargaining Power of Suppliers	Moderate to High	Primary suppliers are LLM API providers. The kernel violation pattern and the cross-platform κ divergence demonstrate that no single provider is structurally reliable across all institutional discourse domains. The model-agnostic Python core is the

Force	Intensity	Analysis
		primary mitigation. The SEC baseline temporal shift (Gemini 18/20 on revalidation) confirms that supplier stability is not guaranteed across time.
Bargaining Power of Buyers	Low to Moderate	Buyers currently have no alternative tool offering structural epistemic auditing. The cost of building a competitor (whether through annotation campaigns or ML training) is prohibitive for most market entrants. Academic audiences have low bargaining power because no integrated tool serves their needs. Governance and compliance audiences have low bargaining power for the same reason.
Threat of Substitutes	Moderate	Substitutes include sentiment analyzers, toxicity detectors, readability metrics, fact-checkers, and fallacy detectors. Each measures a fundamentally different property. None was trained on data designed to capture structural epistemic properties of institutional discourse. Substitution risk is moderate because decision-makers may continue accepting inadequate proxies, not because adequate substitutes exist.
Industry Rivalry	Low	No direct competitor offers multi-dimensional epistemic auditing in a single deterministic output. The analytical categories the engine produces (bias classification plus authority-laundering flags plus learning posture plus coordinate placement) are not offered together by any existing tool. The dataset gap that constrains ML-based competitors is itself a structural barrier: MAFALDA's 200 annotated texts establish the current performance ceiling.

16.5 SWOT Analysis

Table 16-2. SWOT Analysis

Category	Items
Strengths	Novel measurement framework; no existing tool offers deterministic multi-layer structural epistemic auditing.

Category	Items
	Gemini achieves κ of 1.000 and F1 of 1.000 on the BLS Validation Corpus classification dimension. GPT o3 achieves perfect sub-layer precision on the BLS corpus (zero false positives on laundering). The 69-text validation workbook with 19 externally annotated ground-truth documents and the 10-violation kernel registry are the field's most complete annotated resources specifically targeting structural institutional discourse auditing. 89 automated pytest assertions passing in v2.0.0. Three-platform deployment (GPT o3, Gemini, Claude) validates the model-agnostic architecture empirically. Institutional backing of NYU and The Digital Forge; faculty mentorship from Dr. Andres Fortino.
Weaknesses	Evasion and ideological gates remain empirically thin outside synthetic texts: one confirmed ideological case (BLS_B06E) and one confirmed evasion case (SemEval Article 1) across all validation corpora. Kernel violations documented on both platforms across three corpora: Layer 3 under-firing is systematic and requires the Python kernel to enforce the co-occurrence logic independently of the LLM. SEC baseline temporal shift (Gemini 18/20 on revalidation) demonstrates that LLM signal estimation is not stable across time. Reference-classification circularity on the non-externally-annotated corpora: Gemini's κ of 1.000 on BLS measures consistency with a reference it helped define. GPT editorial-register sensitivity on SEC produces kernel violations KV-05, KV-06, KV-07, requiring v2.1 framing_density threshold recalibration. Lexicons developed on U.S. institutional text; non-U.S. and multilingual extension not yet validated.
Opportunities	The validation workbook as a domain corpus: expanding it is both a research contribution and a competitive moat. Formal MAFALDA benchmarking (full 200-text dataset) would yield defensible precision-recall scores. SemEval-2023 Task 3's 23-technique taxonomy provides a validated roadmap for laundering taxonomy extension. Distance laundering formalization as a fourth detection flag is under design for v2.2. Growing demand for AI governance tools (EU AI Act, NIST AI RMF, ISO standards) creates institutional pull for domain-specific annotated corpora and deterministic audit instruments.

Category	Items
	Academic publication pathway is well-suited to prototype plus dataset contribution format: the IEEE conference paper was accepted to ICETM 2026.
Threats	The dataset gap that currently limits ML competitors is temporary: well-resourced technology companies with proprietary institutional corpora could close it faster than the academic community. LLM provider API changes could require interface-layer updates (mitigated by the model-agnostic core). Platform stability is not guaranteed: the SEC baseline shift demonstrates that LLM-based tools must include monitoring for drift. JRC or other institutions could release integrated tools or purpose-built institutional discourse corpora before publication of extended findings. Subjectivity in epistemic constructs may attract methodological critique; this is strengthened if independent blind annotation does not confirm the current κ results. Risk that decision-makers continue to accept inadequate proxies (sentiment tools, readability metrics, fact-checkers) because purpose-built structural auditing tools remain unfamiliar.

The SWOT analysis is clear. The project is well-positioned for publication and deployment due to its novel measurement framework, validated performance, and documented failure modes. The primary strategic goal is to create an independent, blind-scored human-annotated reference standard, which will mitigate circularity risks and enhance claims of accuracy. This is the top priority for future work, as outlined in Section 14.

Appendix A. Project Acceptance Document

The signed Project Acceptance Document (Document 07, Parts 1 and 2) is attached as a PDF insert following this page. The document records the sponsor's formal acceptance of the project's seven objectives and the measured results against each. Parts 1 and 2 carry the original signatures obtained at the start and end of the project respectively.

Sponsor: Dr. Andres Fortino, Ph.D.

Sponsoring Unit: The Digital Forge, NYU School of Professional Studies

Student: Yenesey Concepcion (NetID: yac2027)

Degree: Master of Science in Management and Systems

Term: Spring 2026

Objective acceptance summary. Objective 1 (six-layer deterministic architecture): accepted. Objective 2 (70 percent classification match rate): accepted; achieved approximately 75 percent. Objective 3 (100 percent synthetic gate seed validation): accepted; achieved 100 percent on both platforms. Objective 4 (multi-platform deployment): accepted; achieved on three platforms. Objective 5 (validation workbook with MAFALDA-aligned annotations): accepted; workbook v2.1 contains 69 texts with 53 MAFALDA-aligned annotations. Objective 6 (automated test suite with 50+ assertions): accepted; 89 passing assertions. Objective 7 (complete audit reports): accepted; all 69 documents produced complete four-section outputs.

Sponsor's Project Acceptance Document – PART 2

PROJECT RESULTS

STUDENT: Include below the project goal and ALL the objectives in your project, which you detailed in Project Plan section (07-Part1). You may add additional objectives, or add space, as needed.

Planned Start Date: January 13, 2026 **Planned End Date:** May 15, 2026

Actual Start Date: January 13, 2026 **Actual End Date:** April 8, 2026 (sponsor acceptance)

If actuals differ from planned dates, the revised dates (Actual) are accepted by the sponsor if initialed here: Sponsor Initials _____

Project Goal Was the project goal achieved as planned? ☒ Yes ☐ No, Reason missed: _____

If NO, please explain why this is an acceptable deviation. _____ Sponsor Initials AF

Project Objective #1 — Epistemic Framework Development & Architecture Specification (WBS Phase 3, M2b)

Planned deadline: Framework Documentation Submitted (Milestone M2b) — February 27, 2026.
Outcome: Literature review completed across epistemic bias, construal-level theory, and NLP safety domains. Philosophical constructs (Wittgenstein, Popper, Argyris) mapped to measurable linguistic features. MAFALDA-aligned taxonomy designed; EIS scoring logic defined; fuzzy-logic membership functions and thresholds finalized. Theoretical taxonomy document (E_engine_framework_doc.md) produced and submitted.

Did the student's project meet this objective? ☒ Objective #1 has ☐ has not been met. Sponsor Initials AF

If not met, please explain why this is or is not an acceptable deviation:

Project Objective #2 — MVP Prototype: Engine Core Implementation & Kernel Violation Documentation (WBS Phase 4, M3)

Planned deadline: MVP Complete — Sponsor Review Ready (Milestone M3) — February 20, 2026.
Outcome: LLM-as-Judge evaluation architecture designed (Task 4.1). Core engine implemented with nominalization density and agent deletion in mafalda_engine.py (Task 4.2). Structural linguistic extraction module built (Task 4.3). 4-quadrant topological classifier implemented (Task 4.4). EIS scoring output and prompt generation layer wired (Task 4.5). MVP integration testing completed against sample texts (Task 4.6). Kernel Violation Registry documents 10 confirmed violations across both formal platforms, each traceable to layer, document ID, platform, and signal values.

Did the student's project meet this objective? ☒ Objective #2 has ☐ has not been met. Sponsor Initials AF

If not met, please explain why this is or is not an acceptable deviation:

Project Objective #3 — Corpus Development & Model Validation (WBS Phase 8)

Planned deadline: Alignment evaluation complete — April 6, 2026 (Task 8.6). Outcome: Stratified sampling strategy defined across six institutional source domains: SEC regulatory filings, Bureau of Labor Statistics technical releases, Energy and oil-market journalism (WSJ, FT, and syndicated wire reconstructions), synthetic gate-test cases, SemEval-2023 hyperpartisan articles, and the MAFALDA fallacy benchmark (Task 8.1). Validation workbook (v2.1) assembled comprising 69 texts: 67 documents across six sub-corpora plus 2 calibration seeds, including 19 documents with independently produced external ground truth (Task 8.2). Human Annotation Protocol (HAP) binary decision tree rubric drafted (Task 8.3). Inter-rater reliability study conducted; Cohen's κ generated across classification and authority-shielding dimensions (Task 8.4). First engine validation dry run against the full corpus produced precision and recall metrics (Task 8.5). Alignment score evaluated against independent external ground truth on the SemEval (n=7) and MAFALDA (n=12) sub-corpora (Task 8.6).

Did the student's project meet this objective? ☒ Objective #3 has ☐ has not been met. Sponsor Initials _____

If not met, please explain why this is or is not an acceptable deviation:

Project Objective #4 — Multi-LLM Replication & Cross-Platform Consistency Validation (WBS Phase 9)

Planned deadline: Performance optimization complete — April 20, 2026 (Task 9.6); actual completion accelerated to April 8, 2026 ahead of sponsor acceptance. Outcome: Two formal LLM evaluation platforms selected (OpenAI GPT-4 and Google Gemini) with informal third-platform portability verification on Anthropic Claude (Task 9.1). API access and prompt wrappers configured for each platform (Task 9.2). Engine scoring run across both formal LLM spaces on the full 69-text corpus (Task 9.3). Cross-LLM consistency analysis completed; judge architecture validated with inter-model agreement statistics across six corpora (Task 9.4). EIS weights and fuzzy thresholds revised based on revalidation findings (Task 9.5). Performance optimization and edge-case handling completed (Task 9.6).

Did the student's project meet this objective? ☒ Objective #4 has ☐ has not been met. Sponsor Initials *AF*

If not met, please explain why this is or is not an acceptable deviation:

Project Objective #5 — Final Documentation, Research Paper & Project Closeout (WBS Phases 10–11, M4)

Planned deadline: Sponsor Acceptance Document signed April 8, 2026 (Task 10.6); Project Complete / Graduation Milestone M4 — May 15, 2026. Outcome: Applied research paper drafted covering framework, methodology, and results (Task 10.2) and submitted to a peer-reviewed conference venue on April 3, 2026; acceptance is pending at the time of sponsor sign-off. GitHub repository, README, and

code architecture documentation finalized on April 6, 2026 (Task 10.3). Technical documentation (installation and API guide) produced (Task 10.4). Deliverables compiled into Final Report [DOC J] for submission by April 24, 2026 (Task 10.5). Sponsor Project Acceptance Document Part 2 [DOC 07-Pt2] signed April 8, 2026 (Task 10.6). Final sponsor presentation and demo scheduled for April 28, 2026 (Task 11.1). Lessons learned and closure report scheduled for completion by May 1, 2026 (Tasks 11.2–11.3).

Did the student's project meet this objective? ☒ Objective #5 has ☐ has not been met. Sponsor Initials AF

If not met, please explain why this is or is not an acceptable deviation:

PROJECT ACCEPTANCE

Sponsor: Please show your evaluation of student's work, add comments, and sign below.

SPONSOR: Sponsor's evaluation of student's performance on this project:

☒ A ☐ A- ☐ B+ ☐ B ☐ B- ☐ C+ ☐ C ☐ C- ☐ F

Sponsor's Comments:

An excellent execution of the project resulting in a submitted paper to an IEEE conference in May.

☒ Project was completed satisfactorily and is hereby accepted.

☐ Project was completed satisfactorily but did not meet all objectives, as shown above. Nevertheless, the Project is accepted.

☐ Project was NOT completed satisfactorily and is NOT accepted.

Please explain why not accepted in Sponsor's Comments section, above.

Project Sponsor Signature: Andres Fortino Date: 04/08/26

Student Signature: [Signature] Date: 04/08/26

Sponsor's Project Acceptance Document – PART 1

Project Title: The Epistemic Engine: A Deterministic Structural Auditor for Institutional Discourse

Student Name: Yenesey Concepcion

Sponsoring Organization: Digital Forge, New York University School of Professional Studies – Division of Programs in Business

Project Sponsor Name and Title: Andres Fortino, Ph.D., Faculty Advisor

Project Sponsor Contact Information: agf249@nyu.edu

PROJECT PLAN

At project start, show the project goal; the project objectives and related metrics to be used to show successful project completion. Sponsor should sign to indicate agreement.

Project Goal

Design, implement, and validate the Epistemic Engine: a deterministic, multi-layer Python pipeline that classifies the structural epistemic properties of institutional text (regulatory filings, labor-economics releases, commodity-market journalism, and externally annotated argumentation corpora) using lexicon-based signal densities and threshold-based logic, without relying on subjective human annotation. The project spans January 13 – May 15, 2026, across eleven WBS phases, and culminates in a peer-reviewed conference paper submission and full sponsor sign-off.

Objective #1 — Epistemic Framework Development & Architecture Specification (WBS Phase 3, M2b)

Conduct a literature review spanning epistemic bias, construal-level theory, and NLP safety gaps; map philosophical constructs (Wittgenstein, Popper, Argyris) to measurable linguistic features; design the MAFALDA-aligned taxonomy; define the Epistemic Integrity Score (EIS) scoring logic and fuzzy-logic membership functions and thresholds; and produce `E_engine_framework_doc.md` as the formal theoretical taxonomy deliverable. Metric: framework document published and approved; all EIS scoring dimensions and fuzzy threshold values documented with explicit numeric boundaries. Literature review begins January 20, 2026 (Task 3.1); framework document complete by February 24, 2026 (Task 3.6); Milestone M2b: Framework Documentation Submitted — February 27, 2026.

Objective #2 — MVP Prototype: Engine Core Implementation & Kernel Violation Documentation (WBS Phase 4, M3)

Design the LLM-as-Judge evaluation architecture; implement the core engine covering nominalization density and agent deletion (`mafalda_engine.py`); build the structural linguistic extraction module; implement the 4-quadrant topological classifier; wire the EIS scoring output and prompt generation layer; complete MVP integration testing against sample texts; and document every instance in which a platform's own computed signal densities satisfy threshold conditions but the platform fails to fire the corresponding classification or laundering flag. Metric: MVP passes integration testing (Task 4.6); Kernel Violation Registry completed with at least five confirmed violations, each traceable to layer, document ID, platform, and signal values (expected versus reported). Development begins January 27, 2026 (Task

4.1); integration testing complete February 19, 2026 (Task 4.6); Milestone M3: MVP Complete — Sponsor Review Ready, February 20, 2026.

Objective #3 — Corpus Development & Model Validation (WBS Phase 8)

Define a stratified sampling strategy across institutional source domains spanning SEC regulatory filings, Bureau of Labor Statistics technical releases, Energy and oil-market journalism (WSJ, FT, and wire reconstructions), synthetic gate-test cases, SemEval-2023 hyperpartisan articles, and the MAFALDA fallacy benchmark; conduct an inter-rater reliability study and generate Cohen's κ ; run the first engine validation dry run against the annotated and unannotated texts, producing metrics output by the engine for two formal LLM implementations; and evaluate alignment score against independent external ground truth. Metric: Cohen's κ reported from inter-rater study (Task 8.4); precision and recall metrics produced from dry run (Task 8.5); alignment score evaluated across source domains and annotation provenance (Task 8.6). Corpus work begins February 23, 2026 (Task 8.1); inter-rater study complete March 25, 2026 (Task 8.4); dry run complete April 1, 2026 (Task 8.5); alignment evaluation complete April 6, 2026 (Task 8.6).

Objective #4 — Multi-LLM Replication & Cross-Platform Consistency Validation (WBS Phase 9)

Select two formal LLM evaluation spaces (OpenAI GPT-4 and Google Gemini) with informal third-platform portability verification on Anthropic Claude; configure API access and prompt wrappers for each; run engine scoring across the formal LLM spaces on the full corpus; conduct cross-LLM consistency analysis to validate the judge architecture; revise EIS weights and fuzzy thresholds based on validation findings; and complete performance optimization and edge-case handling. Metric: engine scoring completed on both formal LLMs (Task 9.3); cross-LLM consistency analysis report produced with inter-model agreement statistics (Task 9.4); revised EIS weights and thresholds documented (Task 9.5); edge-case handling verified (Task 9.6). LLM selection begins March 23, 2026 (Task 9.1); scoring runs complete April 8, 2026 (Task 9.3); consistency analysis and threshold revision complete April 15, 2026 (Tasks 9.4–9.5); performance optimization complete April 20, 2026 (Task 9.6).

Objective #5 — Final Documentation, Research Paper & Project Closeout (WBS Phases 10–11, M4)

Draft the applied research paper body covering framework, methodology, and results (Task 10.2); finalize the GitHub repository, README, and code architecture documentation (Task 10.3); produce the technical documentation including installation and API guide (Task 10.4); compile all deliverables into the Final Report [DOC J] (Task 10.5); obtain sponsor final sign-off via the Sponsor Project Acceptance Document Part 2 [DOC 07-Pt2] (Task 10.6); deliver the final sponsor presentation and demo (Task 11.1); and produce lessons learned and project closure documentation (Tasks 11.2–11.3). Metric: research paper drafted and submitted to a peer-reviewed conference venue; Final Report compiled and submitted; [DOC 07-Pt2] signed by sponsor; final presentation delivered; closure report complete. Paper draft begins April 2, 2026 (Task 10.2); paper submitted to conference venue April 3, 2026; GitHub repository finalized April 6, 2026 (Task 10.3); sponsor acceptance signed April 8, 2026 (Task 10.6); Final Report submitted April 24, 2026 (Task 10.5); final paper delivered April 28, 2026 (Task 11.1).

Milestone M4: Project Complete / Graduation — May 15, 2026.

I agree with the above planned project goal, project objectives, and related metrics.

Project Sponsor Signature: Andres Fortino Date: 04/08/26

Printed Name: Andres Fortino, Ph.D.

07-Part1 SponsorsProjectAcceptanceDocument V3.3.docx · Epistemic Engine (Concepcion, Jan–May 2026)

This document is the means by which your project sponsor formally agrees that your project has been satisfactorily completed and that it meets the project goal and objectives that were set at the onset of the project. The “PLAN” section is to be prepared at the beginning of the project and the “RESULTS” and “ACCEPTANCE” sections after your project has been completed. Your sponsor should provide input and sign where indicated.

Appendix B. Project Plan, Baseline, and Actual WBS

The project work breakdown structure was organized around seven phases. The baseline plan allocated twenty weeks across the first six phases. The actual schedule ran three weeks over plan, concentrated in Phase 3 (validation workbook development) and Phase 5 (platform interface deployments), with an unplanned Phase 7 added for the revalidation pass.

Table B-1. Project Work Breakdown Structure

WBS	Activity	Baseline	Actual	Variance	Notes
1.0	Literature Research and Conceptual Architecture	Weeks 1-4	Weeks 1-4	0	On plan.
1.1	Philosophy review: Popper, Argyris, Wittgenstein	Weeks 1-2	Weeks 1-2	0	On plan.
1.2	NLP benchmark review: LIWC, General Inquirer, MAFALDA, SemEval	Weeks 2-3	Weeks 2-3	0	On plan.
1.3	Architectural design: six-layer deterministic pipeline	Weeks 3-4	Weeks 4-5	+1 week	Kierkegaard integration extended design review.
2.0	Python Kernel Implementation	Weeks 5-9	Weeks 5-10	+1 week	Threshold calibration more iterative than planned, particularly Layer 5 coordinate logic.
2.1	Layer 0 signal computation (14 densities)	Weeks 5-6	Weeks 5-6	0	On plan.

WBS	Activity	Baseline	Actual	Variance	Notes
2.2	Layer 2 bias gate hierarchy	Week 7	Week 7	0	On plan.
2.3	Layer 3 laundering detection	Week 8	Week 8	0	On plan.
2.4	Layer 5 coordinate spectrograph	Week 9	Weeks 9-10	+1 week	Near-threshold sensitivity required additional calibration.
3.0	Validation Workbook Development	Weeks 8-12	Weeks 9- 12	+1 week	Expanded from 7-document proof to 69-document multi-corpus framework; added externally annotated ground truth (SemEval, MAFALDA).
3.1	Workbook v1.0 to v1.3 (SEC + BLS integration)	Weeks 8-11	Weeks 9-13	+2 weeks	SEC lexicon coverage gaps required mid-phase lexicon expansion.
3.2	Workbook v2.0 (baseline-validation restructure)	Week 12	Weeks 14-15	+3 weeks	Circularity problem identification motivated structural redesign.
3.3	External corpus integration (SemEval, MAFALDA)	Not planned	Week 16	+1 week new scope	Added to address reference-classification circularity.
4.0	Automated Test Suite Construction	Weeks 11-14	Weeks 11-14	0	On plan. 89 assertions implemented and passing at v2.0.0 release.
5.0	Platform Interface Deployments	Weeks 13-18	Weeks 14-19	+1 week	Per-platform prompt engineering larger than planned.
5.1	GPT o3 interface	Weeks 13-14	Weeks 14-15	+1 week	System prompt refinement.

WBS	Activity	Baseline	Actual	Variance	Notes
5.2	Gemini interface	Weeks 15-16	Weeks 16-17	+1 week	Structured output formatting adjustments.
5.3	Claude interface (informal)	Weeks 17-18	Weeks 18-19	+1 week	Cross-platform architectural portability confirmation.
6.0	Validation, Analysis, and Documentation	Weeks 17-20	Weeks 17-20	0	On plan. Complete validation runs across all corpora; IEEE paper drafted.
7.0	Revalidation Pass	Not planned	2026-03-28	Unplanned	Added after Phase 6 raised questions about circularity and temporal stability; produced the 10-violation kernel registry and SEC baseline shift documentation.

Overall schedule variance: six weeks over plan on the core six-phase schedule, plus one unplanned phase. The variance was concentrated in scope expansions that proved correct in retrospect. The validation workbook scope expansion produced the corpus that established the two principal findings. The revalidation pass produced the kernel violation registry and SEC baseline temporal shift documentation, both of which are consequential for the paper's conclusions. Neither scope addition would be removed from the schedule in retrospect.

Appendix C. References and Bibliography

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Appendix D. Kernel Violation Registry

The kernel violation registry documents deviations from the deterministic specification in which an LLM platform correctly computes signal values at Layer 0 but does not execute the Layer 2 or Layer 3 conditional logic faithfully. Ten confirmed violations were documented across three independent corpora during the 2026-03-28 revalidation.

Table D-1. Kernel Violation Registry

ID	Corpus	Doc	Platform	Layer	Condition	Signal Values	Expected	Reported
KV-01	Synthetic	SYN_01	GPT	L3	bureaucratic_masking: lex greater than 0.02 AND ratio greater than 0.02	lex = 0.0556, ratio = 0.0278	true	false
KV-02	Synthetic	SYN_02	GPT	L3	bureaucratic_masking: lex greater than 0.02 AND ratio greater than 0.02	lex = 0.0588, ratio = 0.0294	true	false
KV-03	Synthetic	SYN_05	Gemini	L3	scientism: ratio greater than 0.02 AND ethos greater than 0.02	ratio = 0.080, ethos = 0.050	true	false

ID	Corpus	Doc	Platform	Layer	Condition	Signal Values	Expected	Reported
KV-04	Synthetic	SYN_06	GPT	L3	bureaucratic_masking: lex greater than 0.02 AND ratio greater than 0.02	lex = 0.1220, ratio = 0.0488	true	false
KV-05	SEC	Doc 3.5	GPT	L3	identity_laundering: ethos greater than 0.03	ethos = 0.0188	false	true
KV-06	SEC	Doc 4.5	GPT	L3	identity_laundering: ethos greater than 0.03	ethos = 0.0176	false	true
KV-07	SEC	Doc 8.5	GPT	L2	ideological: normative greater than 0.05 OR universal greater than 0.02	normative = 0.0149, universal = 0.0025	false	true
KV-08	MAFALDA	MAF_03	GPT	L2	moral_framing: framing greater than 0.02	framing = 0.0323	true	false

ID	Corpus	Doc	Platform	Layer	Condition	Signal Values	Expected	Reported
KV-09	MAFALDA	MAF_11	GPT	L2	moral_framing: framing greater than 0.02	framing = 0.0345	true	false
KV-10	Synthetic	SYN_05	Gemini	L3	bureaucratic_masking: lex greater than 0.02 AND ratio greater than 0.02	lex = 0.0500, ratio = 0.0800	true	false

KV-10 records a second Gemini violation on SYN_05 (bureaucratic_masking), separate from the scientism failure in KV-03. Together, they indicate a compound kernel failure on a single synthetic test case, making SYN_05 the most informative document in the corpus. The violation pattern is structured: Layer 2 violations occur only on GPT with editorial-register text, while Layer 3 violations are present on both platforms. The Layer 3 pattern shows that both platforms correctly compute signal densities but fail to evaluate conditional logic. The architectural response isolates all Layer 2 and Layer 3 conditional logic in the Python kernel, independent of the LLM interface layer.

Appendix E. Energy Benchmark Compression Analysis

This appendix documents the coordinate compression analysis across the 10 documents of the Energy/Oil Market validation corpus. Compression is computed per document as the ratio of Gemini's coordinate magnitude to GPT's coordinate magnitude, using the formula $\sqrt{\text{Gemini_X squared} + \text{Gemini_Y squared}}$ divided by $\sqrt{\text{GPT_X squared} + \text{GPT_Y squared}}$. This is the same formula used in Section 3 of the SEC baseline (Mean Intensity compression), where it produces ratios of approximately 230 times on the X axis, 83 times on the Y axis, and 102 times on intensity.

Table E-1. Per-Document Compression Values

Doc	Type	GPT (X, Y)	Gemini (X, Y)	Compression	Notes
ENE_X1	primary (WSJ)	(0.01, -0.02)	(4, 2.5)	211 times	WTI negative-price crash
ENE_X1E	reconstruction	(0.8, -1.2)	(3, 3)	2.9 times	CNN/Bloomberg/EIA reconstruction
ENE_X2	primary (WSJ)	(0.04, -0.04)	(-1.5, 3.5)	67 times	Saudi unilateral cut
ENE_X2E	reconstruction	(-0.9, -1.1)	(-2.5, 2)	2.3 times	NPR/CNN/PBS reconstruction
ENE_X3	primary (FT)	(4.2, -3.9)	(4.5, 1)	0.80 times (inverted)	Brent 80 dollar rally
ENE_X3E	reconstruction	(0.3, -0.2)	(2, 4)	12 times	EIA/CRS reconstruction
ENE_X4	primary (FT)	(0.03, -0.01)	(5.5, 2)	185 times	Negative-price FT analysis

Doc	Type	GPT (X, Y)	Gemini (X, Y)	Compression	Notes
ENE_X4E	reconstruction	(1.2, -0.4)	(3.5, 3)	3.6 times	Bloomberg/CNBC reconstruction
ENE_X5	primary (FT)	(3.8, 0.9)	(-1, 2.5)	0.69 times (inverted)	OPEC+ voluntary cuts
ENE_X5E	reconstruction	(2, -1.5)	(2.5, -2)	1.3 times	NPR/AP/CNN OPEC+ surprise cuts

Three patterns emerge from Table E-1.

First, WSJ primary reporting (X1, X2, X4) exhibits extreme compression (67 to 211 times) matching the SEC baseline magnitude of approximately 230 times. This confirms that GPT's coordinate estimation on polished institutional primary journalism is systematically compressed relative to Gemini's, consistent with the SEC baseline pattern.

Second, FT articles X3 and X5 show inverted compression (GPT's coordinate magnitude exceeds Gemini's). This inversion corresponds exactly to the two documents in the Disagreement Decomposition where GPT over-fires Gate 2 or Gate 3: X3 triggers moral_framing on GPT due to geopolitical conflict language and X5 triggers ideological on GPT due to prescriptive market rhetoric. The coordinate expansion and the gate misfire are not independent errors. They share a cause in GPT's sensitivity to editorial and geopolitical vocabulary, which simultaneously inflates HEAD and DISRUPTION

densities above Gemini's readings and pushes framing or normative densities past their gate thresholds.

Third, reconstructions cluster at 2.3 to 3.6 times compression across X1E, X2E, X4E, matching the approximate 3 times characterization in the published IEEE paper.

ENE_X5E at 1.3 times compression (the only reconstruction where both platforms converge on the same non-clean classification, moral_framing, with matching Q4 quadrant signs) represents the single Energy document where model outputs approach alignment. X3E at 12 times compression is the sole exception among reconstructions, driven by GPT's near-zero output on that specific document.

The compression analysis provides a quantified cross-platform measurement of GPT's coordinate compression on institutional primary journalism. It is one of the components of the register-mismatch gradient documented in Section 8: the structural signal available to Layer 5 differs across platforms in ways that track the content and register of the document under audit.